

MULTILINGUAL CONVERSATIONAL AI FOR LABOUR FORCE SURVEYS: A MULTI-AGENT RAG SYSTEM WITH CREWAI FRAMEWORK

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DECLARATION

I, **Sivarama Krishnan Subbiah (Roll No: 2024MCS110007)**, hereby declare that this report entitled “**Multilingual Conversational AI for Labour Force Surveys: A Multi-Agent RAG System with CrewAI Framework**” submitted to Indian Institute of Information Technology Kottayam towards partial requirement of **Master of Technology in Artificial Intelligence and Data Science** is an original work carried out by me under the supervision of **Dr. Goutam Mali** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. External information, methods, and results used in this report have been acknowledged and cited. The reported experiments are distinguished from planned studies and unverified claims.

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CERTIFICATE

This is to certify that the work contained in this project report entitled “**Multilingual Conversational AI for Labour Force Surveys: A Multi-Agent RAG System with CrewAI Framework**” submitted by **Sivarama Krishnan Subbiah** (Roll No: **2024MCS110007**) to Indian Institute of Information Technology Kottayam towards partial requirement of **Master of Technology** in **Artificial Intelligence and Data Science** has been carried out by him under my supervision and that it has not been submitted elsewhere for the award of any degree.

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ABSTRACT

Labour Force Surveys require consistent measurement of employment and standardised coding of occupations, industries, and education. Multilingual respondent descriptions and ambiguous job titles make automated coding difficult. This thesis develops and evaluates a conversational prototype targeting English, Arabic, Urdu, Hindi, and Tagalog, with dictionary-based Gulf Arabic normalisation and a modular CrewAI-assisted backend.

The system combines a deterministic interview state machine with retrieval-based ISCO-08 classification, ISIC Rev. 4 industry classification, and ISCED 2011 attainment and ISCED-F 2013 field mapping. A Semantic Relation Engine checks occupation–industry and occupation–attainment compatibility using disclosed, project-authored rules. Its coherence score is a heuristic, and HIGH violations trigger human-review queue creation. Rule and endpoint tests establish agreement with implementation expectations; independent expert validation and probability calibration remain outstanding.

On a project-generated, occupation-group-separated WISCO benchmark of 18,747 language-specific reference titles, flat retrieval achieved 21.19% exact unit-group accuracy compared with 10.35% for hierarchical retrieval. Combining official-definition enrichment with a larger intended multilingual encoder produced the highest observed accuracy, 40.95%; that run’s preserved records do not establish which encoder actually executed, so the gain is not attributed to encoder size. The configuration the prototype actually serves is a fragment-to-parent retriever reaching 38.83% on the same records, 6.28 points above its own same-encoder baseline and bound to a recorded serving identity. These later comparisons reused the same benchmark during development and are therefore interpreted as comparative evidence requiring confirmation on an untouched test set. Tested LLM reranking configurations did not show consistent additional gains. Industry and education-field experiments use synthetic data and are reported separately from occupation-reference data and human survey evidence.

The thesis contributes a comparison of retrieval designs with preserved predictions, an explicit cross-standard review mechanism, and an analysis of the prototype’s empirical and operational limitations. A synthetic 30-scenario interview run demonstrates execution of the implemented workflow, but does not measure respondent usability or burden reduction. The planned human pilot has not been conducted and requires ethics approval before recruitment. Routing, persistence, and access-control defects identified in review have been repaired and regression-tested, with re-verification in the deployment environment still outstanding. The results support further research on multilingual classification and assisted survey coding rather than immediate replacement of a validated field survey instrument.

Keywords: Labour Force Survey; conversational AI; multi-agent systems; CrewAI; retrieval-augmented generation; ISCO-08; ISIC Rev.4; ISCED-F 2013; Semantic Relation Engine; multilingual NLP; code-switching; human-in-the-loop; official statistics.

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Chapter 1

Introduction

Labour Force Surveys (LFS) provide information on employment, unemployment, and labour underutilisation. Their value depends on the consistent application of statistical concepts and on the quality of the responses collected. The International Labour Organization (ILO) establishes the relevant conceptual framework through the International Conference of Labour Statisticians [9]. Within that framework, classifying open descriptions of work is a measurement task: the wording supplied by a respondent must be interpreted in relation to a defined coding scheme.

This thesis investigates a UAE-oriented conversational LFS prototype. The implemented questionnaire includes employment, education, demographic, and working-condition items, together with UAE-specific fields such as emirate and residence duration. The interface provides English, Arabic, Urdu, Hindi, and Tagalog, with additional processing for Gulf Arabic. The prototype integrates occupation coding to ISCO-08, industry coding to ISIC Rev. 4, and education coding to ISCED 2011 and ISCED-F 2013. These are the classification versions selected for this study; the prototype is not an authorised national survey instrument.

The central problem is how to connect flexible language input with reproducible classification and questionnaire control. Conversational fluency alone does not establish correct coding, and successful software execution does not establish valid measurement of a population. The research therefore separates classification effectiveness, cross-standard checks, language robustness, and operational feasibility.

1.1 Motivation and Background

1.1.1 Occupation Coding as a Measurement Problem

ISCO-08 groups jobs according to the tasks and duties performed. Its four-level structure contains 10 major groups, 43 sub-major groups, 130 minor groups, and 436 unit groups [8]. A description such as *I repair equipment* may fit several unit groups unless the equipment, tasks, or work setting is specified. A classifier must therefore handle incomplete descriptions

and distinguish related occupations at different levels of detail.

Coding during an interview is an established research direction. Schierholz et al. [13] investigated supervised candidate suggestions that respondents could select while an interview was in progress. More recently, SOCbot combines retrieved occupation candidates with LLM-based classification and dynamic follow-up questions [14]. These precedents motivate a study of retrieval configurations and their integration into a broader survey; they preclude treating real-time occupation coding or conversational probing alone as a new contribution.

1.1.2 Related but Distinct Classification Targets

Occupation, industry, and education describe different aspects of a respondent’s situation. ISCO-08 concerns work tasks; ISIC classifies the economic activity of the relevant establishment; ISCED describes education programmes and attainment, while ISCED-F describes fields of education and training [8, 16–18]. For example, a software developer can work in a hospital without the occupation and industry classifications being inconsistent. Similarly, formal qualifications are not a complete description of the skills acquired through experience.

Integrating these targets creates an opportunity for review of unusual combinations, but it also creates a risk of rejecting legitimate cases. This thesis consequently treats its Semantic Relation Engine (SRE) as a heuristic screening mechanism. Its hand-built associations and severity rules identify cases for review; they are not an official correspondence table or proof that a respondent’s answers are incorrect.

1.1.3 Multilingual Input and Arabic Dialect Variation

Arabic language processing involves Modern Standard Arabic, regional varieties, and code-switching with other languages. Hamed et al. [7] review the resulting resource and modelling challenges. In a survey, these challenges occur together with specialised occupation terminology and short answers. Translating the questionnaire is therefore only one part of providing language support: answer interpretation, routing, and classification must also preserve the meaning of each response.

The prototype uses multilingual embeddings and a dictionary of Gulf Arabic normalisation rules. Its evaluation distinguishes the availability of language-specific interface text from measured performance on labelled examples. Neither multilingual model training nor a translated interface establishes equivalent accuracy across the five intended languages.

1.1.4 Questionnaire Control and Operational Feasibility

Computer-assisted personal interviewing (CAPI) already supports automated skip logic and validation. For example, World Bank Survey Solutions evaluates enabling and validation conditions and supports multilingual questionnaires [22, 23]. The proposed system extends these

familiar controls with interpretation of conversational responses and classification during the same workflow. Its contribution is evaluated as an integration of components rather than as the invention of conditional routing or real-time validation.

The prototype also reuses stored responses through a Person Register, records classification decisions, and provides human-in-the-loop (HITL) review. These mechanisms can be tested for functional correctness. Claims about reduced respondent burden, interview cost, or improved participation require a comparison with human respondents and an appropriate survey administration baseline. Such effects cannot be inferred from question counts or automated processing time alone.

Table 1.1: *Research problems and the evidence required to assess them.*

Problem	Design response	Required evidence
Ambiguous occupation descriptions	Catalogue retrieval and optional reranking	Heldout coding accuracy and error analysis
Related classification targets	Separate classifiers and explicit coherence rules	Coverage, detection errors, and review behaviour
Language and dialect variation	Multilingual input processing and language-specific answer handling	Per-language evaluation and normalisation ablations
Conversational survey administration	Controlled interview flow, persistence, and HITL review	Functional checks and operational measurements; human studies for burden and acceptability

1.2 Research Approach

The system combines a finite-state conversation controller with directly invoked processing modules. CrewAI provides agent abstractions for some modules, while questionnaire transitions and module invocation remain under application control. The implementation does not use autonomous hierarchical delegation. This architecture makes the routing rules and the points at which classification and review occur inspectable.

Retrieval is studied as a configurable classification component. A flat configuration searches the unit-group catalogue directly; a hierarchical configuration restricts successive searches using predicted parent groups. Catalogue enrichment and embedding size provide additional experimental factors. Multilingual E5 supplies model sizes with different resource requirements [19]. The operational default and the best-performing benchmark configuration are kept distinct when interpreting the results.

The evaluation uses three forms of evidence: external labelled occupation data, synthetic test cases, and software and operational checks. The external WISCO dataset supports occupation-classification comparisons [15]; synthetic cases support narrower studies of classifiers, coherence rules, and conversation execution. These evidence sources address different questions and are not pooled as if they represented the same population or validation setting.

1.3 Research Questions

- RQ1. Retrieval effectiveness.** How do flat and hierarchical retrieval, catalogue enrichment, and embedding choice affect ISCO-08 classification effectiveness on an external held-out benchmark?
- RQ2. Classification integration and review.** How can occupation, industry, and education coding be integrated with transparent cross-standard screening, and what evidence supports the resulting classifications and escalation rules?
- RQ3. Language robustness.** What do the available language-specific evaluations establish about multilingual and dialectal input, and does dictionary-based Gulf Arabic normalisation improve classification on the tested sample?
- RQ4. Operational feasibility.** To what extent can the modular conversation system execute its interview paths, validation, persistence, and review workflow reliably, and which claims still require evaluation with human respondents?

1.4 Research Objectives

The aim is to design and evaluate an integrated conversational survey prototype while establishing the limits of its evidence. Table 1.2 operationalises the research questions. The measures describe how each objective is assessed, rather than postulated accuracy or cost targets. Chapter 6 reports which assessments were completed and where evidence remains incomplete.

1.5 Research Contributions

C1 — Comparative ISCO-08 retrieval evidence. The study compares flat and hierarchical retrieval under documented configurations on an external benchmark. It also evaluates catalogue enrichment and embedding choice. Within the evaluated setting, flat retrieval outperforms the hierarchical configuration. This finding is task- and configuration-specific; it is not a general rejection of hierarchical information retrieval.

Table 1.2: *Objectives, research questions, and assessment measures.*

ID	Objective	Assessment
O1	Implement a controlled conversational questionnaire (RQ4)	Interview-path and transition correctness; response preservation; completed and failed synthetic sessions
O2	Compare ISCO-08 retrieval configurations (RQ1)	Exact unit-group and higher-level accuracy, uncertainty intervals, paired configuration comparisons, and error analysis
O3	Implement ISIC Rev. 4 industry coding (RQ2)	Catalogue coverage and class-level performance on the available synthetic evaluation; explicit treatment of uncovered classes
O4	Implement education-attainment and field coding (RQ2)	Separate evaluation of attainment level and detailed field; catalogue coverage and synthetic-data limitations
O5	Implement auditable coherence screening and escalation (RQ2)	Rule-level behaviour, available detection-error analysis, and verification that HIGH-severity cases enter the review workflow
O6	Assess language and dialect handling (RQ3)	Results by the languages represented in each dataset; normalisation ablation; disclosure of missing native-speaker validation
O7	Integrate pre-fill and response validation (RQ4)	Preservation and correction of stored fields; deterministic consistency-rule checks on relevant paths
O8	Support traceable operation and reporting (RQ4)	Stored method and review records, report generation, processing time, resource constraints, and deployment checks

C2 — Transparent cross-standard screening. The SRE combines hand-built occupation–industry and occupation–education associations with explicit scoring and severity rules. It provides an inspectable mechanism for identifying cases for HITL review. Its scores are heuristic indicators, not empirically calibrated probabilities of correctness; expert validation and assessment of legitimate atypical cases remain necessary.

C3 — An integrated conversational survey prototype. The implementation connects questionnaire routing, multilingual input, multiple classification targets, deterministic validation, persistence, and review in one workflow. This is a system integration contribution. It does not establish that these individual techniques are new or that multi-agent orchestration itself improves accuracy.

C4 — Evaluation and diagnostic artifacts. The project provides configuration-specific benchmark results, synthetic evaluation cases, functional checks, and operational records. These artifacts expose coverage limitations and negative results, including the absence of a measured benefit from Gulf normalisation in the tested sample. They support inspection and subsequent replication without substituting synthetic evidence for respondent validation.

1.6 Scope and Limitations

The scope is a text-based research prototype with UAE-oriented fields, not a representative national LFS or an approved replacement for an official collection system. Sampling design, population weighting, and estimation of national labour-market indicators are outside the study. The implemented questionnaire paths and rules require further review before claims of full compliance with official survey standards can be made.

External occupation evidence comes from WISCO and does not represent the intended UAE respondent population or natural conversational responses in the five interface languages. ISIC and ISCED-F evaluation relies on taxonomy-grounded synthetic data, and their indexed catalogues have incomplete coverage. Coverage also varies by catalogue profile because non-standard legacy entries are excluded from enriched collections. Chapter 3 reports the relevant inventories, and Chapter 5 specifies the denominators and applicability of each evaluation.

The Person Register reuses records from this application’s database. Integration with a national identity or administrative register is outside the scope. Functional checks of pre-fill do not measure respondents’ willingness to confirm stored answers or the resulting measurement error. Similarly, classification confidence thresholds and SRE severity rules have not been established as calibrated measures of risk on representative respondent data.

A separate synthetic operational run includes 30 scripted English interviews. It measures system execution and processing time, not human completion time, satisfaction, or participation. The planned randomised study with human respondents was not conducted. Accordingly, this thesis does not establish reductions in respondent burden or collection cost,

non-inferiority to interviewer administration, or population-level fairness. Ethics approval, recruitment, native-speaker review, independent coding, and an appropriately justified analysis plan are prerequisites for those assessments.

1.7 Thesis Organisation

Chapter 2 reviews occupation coding, retrieval, multilingual processing, and survey-system integration, then maps the research gaps to the questions above. Chapter 3 presents the system architecture, questionnaire controller, classifiers, and coherence rules. Chapter 4 describes the implementation and functional checks. Chapter 5 specifies the datasets, comparisons, statistical methods, and the proposed human-study protocol. Chapter 6 reports the measured outcomes and their validity limits. Chapter 7 discusses the contributions, remaining gaps, and directions for further research.

Chapter 2

Related Work and Literature Review

This chapter develops a focused narrative review of research relevant to the prototype: occupation measurement and coding, retrieval methods, multilingual processing, and controlled conversational workflows. Sources are selected for their relationship to the classification task or survey design. Official classification manuals define the target concepts; research papers provide methods and empirical evidence; software documentation establishes existing platform capabilities. Working papers and technical reports are distinguished from published journal and conference studies. The review does not constitute an exhaustive systematic review or a pooled comparison of published accuracy estimates.

2.1 Labour Force Surveys and Occupation Coding

2.1.1 Classification and Measurement

ISCO-08 defines occupations using the similarity of work tasks and duties and organises them through major, sub-major, minor, and unit groups [8]. This structure permits analysis at different levels of detail, but makes four-digit coding sensitive to incomplete descriptions. The skill model considers tasks, formal education, and training or experience. Education is therefore relevant contextual information, rather than a sufficient rule for assigning or rejecting an individual's occupation.

Industry and education require separate measurement decisions. ISIC Rev. 4 concerns productive activities of statistical units, whereas ISCED 2011 and ISCED-F 2013 describe education levels and fields [16–18]. Neither a job title nor an employer's broad sector necessarily supplies enough information for detailed industry coding. For an integrated survey, target-specific questions and explicit treatment of missing information are therefore as important as the classifier used.

2.1.2 Coding During the Interview

Schierholz et al. [13] studied occupation coding within a telephone interview using supervised candidate suggestions and respondent selection. Their IAB discussion paper evaluates the approach with the German KldB classification and discusses interview duration and coding quality. It demonstrates that adaptive interview coding predates LLM-based systems. Its results concern a different taxonomy and administration mode, so they do not provide a numerical baseline for this thesis’s ISCO-08 retrieval experiment.

SOCbot, presented by Sturgis et al. [14], combines retrieval of UK SOC20 candidates, LLM classification, and follow-up questions when the initial response is insufficient. The published study includes static classification and a dynamic survey implementation. Its comparisons with a production coding pipeline assess coder agreement; the authors distinguish this from accuracy against an independently known true occupation. SOCbot is consequently close prior work for retrieval-assisted conversational occupation measurement. This thesis extends the application setting to an integrated UAE-oriented prototype with additional classification targets, while conducting a separate comparison of ISCO retrieval designs. It does not claim priority for combining RAG and interview probing.

2.1.3 LLM-Based Occupation Standardisation

Li et al. [11] introduce LLM4Jobs for unsupervised occupation extraction and standardisation from free-text job descriptions. The approach is evaluated on synthetic and real-world datasets at different coding granularities. It establishes that occupation coding can be studied as a language understanding and standardisation task without fine-tuning a supervised classifier for every target category. Its classification experiments do not establish the quality of a complete LFS interview or the effect of adaptive questioning on respondents.

The IEA report by Duckworth and Fraillon [4] fine-tunes XLM-RoBERTa for parental occupation coding using ICILS responses and ILO occupation titles. It compares multilingual and language-specific models and uses hierarchical evaluation measures to recognise partially correct classifications. Performance differs when a model is evaluated on its source data and when it is evaluated on the ICILS response data, illustrating the importance of the evaluation distribution. Hierarchical evaluation here concerns partial credit for predictions, rather than the parent-filtered retrieval algorithm investigated in this thesis. The report is relevant to short occupation responses and multilingual coding, although its population and collection context differ from an LFS. Its results and those of LLM4Jobs motivate methodological comparisons; they do not establish performance on this prototype’s target population.

The approaches in Table 2.1 use different taxonomies, populations, input information, and reference-coding procedures. Ranking them by headline accuracy would conflate these differences. A direct superiority claim would require running comparable implementations on a shared, independently labelled dataset with a common protocol.

Table 2.1: *Selected prior approaches and their relevance to this study.*

Source	Relevant approach	Comparison boundary
Schierholz et al. [13]	Supervised candidate suggestions during an interview	German KldB and telephone administration; not the WISCO protocol
Li et al. [11]	Unsupervised LLM occupation extraction and standardisation	Coding experiments do not evaluate this integrated survey workflow
Sturgis et al. [14]	Retrieved candidates, LLM coding, and dynamic survey probes	UK SOC20; coder agreement and survey outcomes are distinct from heldout ISCO accuracy
Duckworth and Fraillon [4]	AI-assisted parental occupation coding	International education-study context
This thesis	Flat versus hierarchical ISCO retrieval and integrated classification and review	External occupation benchmark; synthetic evidence for other targets and operation

2.2 Retrieval-Augmented Methods for Classification

2.2.1 Retrieval and Grounding

Lewis et al. [10] introduced RAG by combining retrieved external information with a generative language model for knowledge-intensive NLP tasks. Retrieval makes relevant information available to a model, but the existence of retrieved context alone does not guarantee that the final answer is correct. For occupation coding, retrieval can instead be used to propose entries from a fixed catalogue, with a reranker or deterministic selection step choosing a code.

This distinction matters for the present implementation. Embedding-based catalogue retrieval can produce a classification without invoking a generative model, while optional LLM reranking selects among candidates. These stages require separate measurement: retrieval recall determines whether the reference code is available for selection, and final accuracy determines whether it is selected. An explanation attached to a code is not itself evidence of correct classification.

Gao et al. [5] review retrieval and generation components and their evaluation. The modular view supports comparing retrieval design, catalogue content, and reranking rather than treating the whole pipeline as a single undifferentiated RAG method.

2.2.2 Flat and Hierarchical Retrieval

A taxonomy suggests a natural hierarchical search strategy: select a major group and restrict subsequent retrieval to its descendants. This can narrow the candidate space, but an incor-

rect parent decision can exclude the correct unit group. Flat retrieval avoids that particular restriction by searching unit groups directly. The relative performance of the two depends on catalogue representations, candidate budgets, query information, and the data distribution.

Du et al. [3] investigate hierarchical retrieval interfaces for agentic question answering. Their A-RAG framework exposes keyword search, semantic search, and chunk reading to a model at different granularities. Hierarchy in that setting concerns retrieval tools and document access; it differs from predicting successive occupation groups and filtering descendants. Its question-answering findings therefore do not establish that a parent-filtered ISCO classifier should outperform flat unit-group retrieval.

The present study addresses this task-specific uncertainty with matched configuration comparisons on WISCO [15]. Its conclusion about flat and hierarchical retrieval is restricted to the evaluated implementations and dataset. Additional ablations of catalogue enrichment, embedding size, and reranking establish which changes have supporting evidence and which remain configuration choices.

2.2.3 Multilingual Embedding Models

Wang et al. [19] describe the multilingual E5 family, including small, base, and large models. Their training and evaluation demonstrate a practical route to multilingual text representations with different resource requirements. Such models permit descriptions in several languages to be compared with catalogue text, but generic embedding benchmarks are not occupation-coding benchmarks. In this thesis, embedding size is consequently evaluated on the classification task and considered together with memory and latency constraints.

2.3 Multilingual NLP and Code-Switching

Hamed et al. [7] review Arabic code-switching across language varieties and between Arabic and other languages, including limitations in available data and modelling approaches. Their review motivates attention to dialectal and mixed-language inputs; it does not establish how frequently these occur in this prototype’s target respondent population or that a particular normalisation method improves survey coding.

The prototype’s Gulf Arabic dictionary is a lightweight preprocessing choice. Replacing a recognised dialect token can make a description closer to the catalogue wording, but it can also leave unrecognised variants unchanged or alter useful information. The relevant empirical question is whether normalisation changes downstream classification under a controlled comparison. The reported ablation uses generated dialectal text, so its outcome cannot establish performance on spontaneous speech, noisy transcription, or the full range of Gulf varieties.

Language support also requires consistent interpretation of categorical answers, negation, numbers, and field corrections. A translated question can be displayed correctly while its answer is interpreted incorrectly. Functional routing tests therefore complement classifier benchmarks, and native-speaker review and cognitive interviewing remain necessary for measurement equivalence. The WISCO languages, the synthetic evaluation languages, and the five interface languages are reported as separate coverage dimensions in Chapter 5.

2.4 Conversational Workflow and Quality Control

2.4.1 Existing Computer-Assisted Survey Capabilities

Survey Solutions implements conditional question enabling, validation expressions, and multilingual questionnaires [22, 23]. These are established capabilities of computer-assisted survey systems. An appropriate comparison for a conversational prototype must therefore consider a well-programmed electronic questionnaire, rather than assume that all conventional instruments lack routing or consistency checks.

The additional design problem is preserving those controls when an answer is obtained in free text and can update multiple fields. The prototype uses a finite-state controller for interview progression and separates field interpretation from classification and validation. Tests of these boundaries assess software behaviour; a field study must assess whether the same design improves or worsens response quality in use.

2.4.2 Modular and Multi-Agent Orchestration

Guo et al. [6] review LLM-based multi-agent systems and their applications and challenges. Multiple specialised components can organise responsibilities, but the framework label alone does not establish a benefit over a simpler pipeline. Relevant comparisons must hold data, models, and task definitions sufficiently constant to isolate the effect of orchestration.

The current system uses CrewAI abstractions [2] within an application-controlled workflow. Modules are invoked directly, with delegation disabled, rather than through an autonomous hierarchical manager. The thesis therefore evaluates module invocation and data flow as engineering properties. It does not attribute classification gains to multi-agent collaboration without a corresponding controlled ablation.

2.4.3 Cross-Standard Screening and Human Review

Because occupation, industry, and education describe related but distinct concepts, cross-standard associations can help identify records that deserve examination. Such associations also reflect assumptions about typical work and education patterns. An atypical combination may represent an error, a valid career transition, an unusual establishment, or skills acquired

outside formal education. A screening score must consequently be distinguished from a truth label.

The proposed SRE uses explicit, hand-built associations and deterministic severity rules. Its evaluation can establish whether a rule fires on its test cases and whether flagged cases enter the review queue. Establishing its value for official statistics additionally requires independent expert annotation, estimates of false positive and false negative rates, and analysis of reviewer workload. Authentication and audit logging support traceability but do not by themselves demonstrate privacy compliance or production security.

2.5 Research Gaps and Position of the Thesis

The reviewed work supplies precedents for automated occupation coding, retrieval-assisted probing, multilingual representations, and survey validation. The gaps below are scoped empirical and integration questions for this prototype, rather than claims that these capabilities are absent from all previous systems.

- G1. Task-specific retrieval evidence.** The existence of a classification hierarchy does not determine the best retrieval strategy. A controlled ISCO comparison is required to distinguish hierarchy, embedding, and catalogue effects (RQ1).
- G2. Evidence for integrated classification and screening.** A system that produces occupation, industry, and education codes must establish their coverage and the behaviour of review rules separately. Independent validation of those associations remains necessary (RQ2).
- G3. Transfer to intended languages and dialects.** Multilingual embeddings and interface translations leave uncertainty about answer interpretation, classification, and measurement equivalence in the intended survey population (RQ3).
- G4. From operational execution to respondent outcomes.** Functional checks and synthetic interviews assess system behaviour; they do not establish burden, satisfaction, cost, or survey-mode effects. The transition to a human study remains an evidence gap (RQ4).

The strongest comparative evidence concerns ISCO retrieval on an external benchmark. The other components have narrower functional or synthetic support. This positioning preserves the distinction between implemented capabilities, measured outcomes, and unanswered research questions used throughout the evaluation and conclusion chapters.

Table 2.2: *Mapping research gaps to objectives, available evidence, and open work.*

Gap and objectives	Evidence in this thesis	Remaining validation
G1: O2	External WISCO configuration comparisons, accuracy, and ablations	Independent replication and transfer to UAE descriptions; direct shared-data comparisons with prior systems
G2: O3–O5	Synthetic classifier evaluations, catalogue audit, and rule/escalation checks	Wider catalogue coverage; real jointly labelled records; expert review of associations and detection errors
G3: O6	Dataset-specific language results and a generated Gulf normalisation ablation	Native-speaker and cognitive review; naturalistic dialect and code-switch samples; subgroup error analysis
G4: O1, O7, O8	Interview-flow checks and scripted operational execution	Ethics-approved human evaluation; independent coding; justified sample size; respondent and review-workload outcomes

Chapter 3

Proposed System Architecture

The prototype combines a multilingual interview interface, deterministic questionnaire control, retrieval-based classification, and human-review support. This chapter distinguishes the survey API workflow from alternative configurations evaluated offline.

3.1 Design Principles

- P1. Traceability.** Method labels, confidence scores, candidate information, configuration identifiers, and logs support inspection. Their correctness requires verification; their presence alone does not establish reproducibility.
- P2. Taxonomy alignment.** Classification references ISCO-08, ISIC Rev. 4, ISCED 2011, and ISCED-F 2013 [8, 16–18]. Coverage and code-normalisation exceptions are reported separately.
- P3. Privacy-aware design.** Authentication, respondent-session ownership checks, audit records, and soft deletion are implemented. Authorisation and data-lifecycle defects identified in the October 2026 review have since been repaired and regression-tested (Chapter 4); verified repairs do not by themselves establish deployment readiness or regulatory compliance.
- P4. Explicit uncertainty.** Retrieval similarities and rule scores support review decisions. They are heuristics, not empirically calibrated probabilities of correct classification.
- P5. Deterministic control.** Field order, clarification, skip conditions, confirmation, and cross-answer validation are distinct from classification and semantic-coherence checks.
- P6. Modularity.** The live API invokes component modules directly, without CrewAI hierarchical delegation. The repository also contains a separate experimental hierarchical Crew, outside this default workflow.

3.2 High-Level System Architecture

Interface layer. Next.js provides login, interview, report, and supervisor-review pages. Prompts target English, Arabic, Urdu, Hindi, and Tagalog, with Gulf Arabic handled as an Arabic variety. Translated interfaces and right-to-left presentation do not establish correct language-specific answer handling. The Hindi/Urdu routing defect found in that review, and its repair, are described in Chapter 4.

Application layer. FastAPI authenticates requests, checks session ownership, invokes components, persists answers, and assembles responses. CrewAI constructs role-specific agents and tasks, while field routing, catalogue lookup, and core coherence scoring are deterministic. Counting agent-constructing modules does not measure agents executed per turn.

Knowledge and persistence layer. Qdrant collections are identified by catalogue/embedding profile. PostgreSQL holds users, OTPs, sessions, responses, audit/access/decision logs, quality reviews, HITL entries, reports, and person-register records. Redis stores conversation state with a 24-hour default time to live. An in-process context cache serves active sessions, with best-effort Redis restoration after restart. Database, cache, and Redis updates are not one atomic transaction.

Model-access layer. The LLM client exposes `GENERAL` and `CRITICAL`. General tasks default to Ollama at temperature 0.3 and may try configured Anthropic, Gemini, Groq, and OpenRouter providers. Critical tasks select the configured Claude model at temperature 0.0 without provider fallback. Unpinned ISCO reranking requests `GENERAL`; validation and quality-report components request `CRITICAL`. Evaluation uses a strict model-selection function where a model is explicitly pinned. Call-site configuration and execution records determine which model ran. Successful end-to-end Claude inference is not established by the archived experiments.

3.3 Agent Module Specification

Table 3.1 summarises the components. Input, state, availability, and optional flags determine execution on each turn.

Table 3.1: *Main components and their implemented responsibilities.*

Component	Responsibility	Execution
Language processor	Language/script detection, Gulf normalisation, code-switch detection, entity extraction	Rules and optional LLM
Conversation manager	Field extraction, state control, clarification, response generation	Rules; LLM or template reply

Table 3.1 — continued

Component	Responsibility	Execution
ISCO classifier	Occupation retrieval and optional reranking	Configured retrieval; legacy fallbacks
ISIC classifier	Industry classification	Keyword/LLM default; optional retrieval
ISCED classifier	Attainment and field-of-study classification	Attainment rules; keyword/retrieval field mapping
Semantic relation engine	Occupation–industry and occupation–attainment checks	Deterministic live core
Cross-standard coordinator	Reconsider an uncertain occupation using cross-standard signals	Optional deterministic flag
Validation agent	Ten cross-answer rules and semantic checking	Rules plus available critical-tier LLM
Quality manager	Completeness/confidence metrics and quality reviews	Rules; optional narrative
Context memory	State persistence and restoration	Redis/cache
Audit logger	Interaction, access, and decision records	Database logging
Report generator	Profile aggregation, classification enrichment, bilingual narrative, caching	Aggregation; optional LLM
Emotional-intelligence component	Engagement and support signals	Heuristic LLM; skipped on some turns
Nationality/person register	Nationality coding and returning-user data	Deterministic services
Query planner, RAG expert, hierarchical Crew	Alternative planning, retrieval, and delegation	Experimental or explicitly enabled paths

3.4 Conversation Flow

The five states are GREETING, COLLECTING_INFO, CLARIFYING, VALIDATING, and COMPLETING. A greeting advances to collection; failure to populate the requested field en-

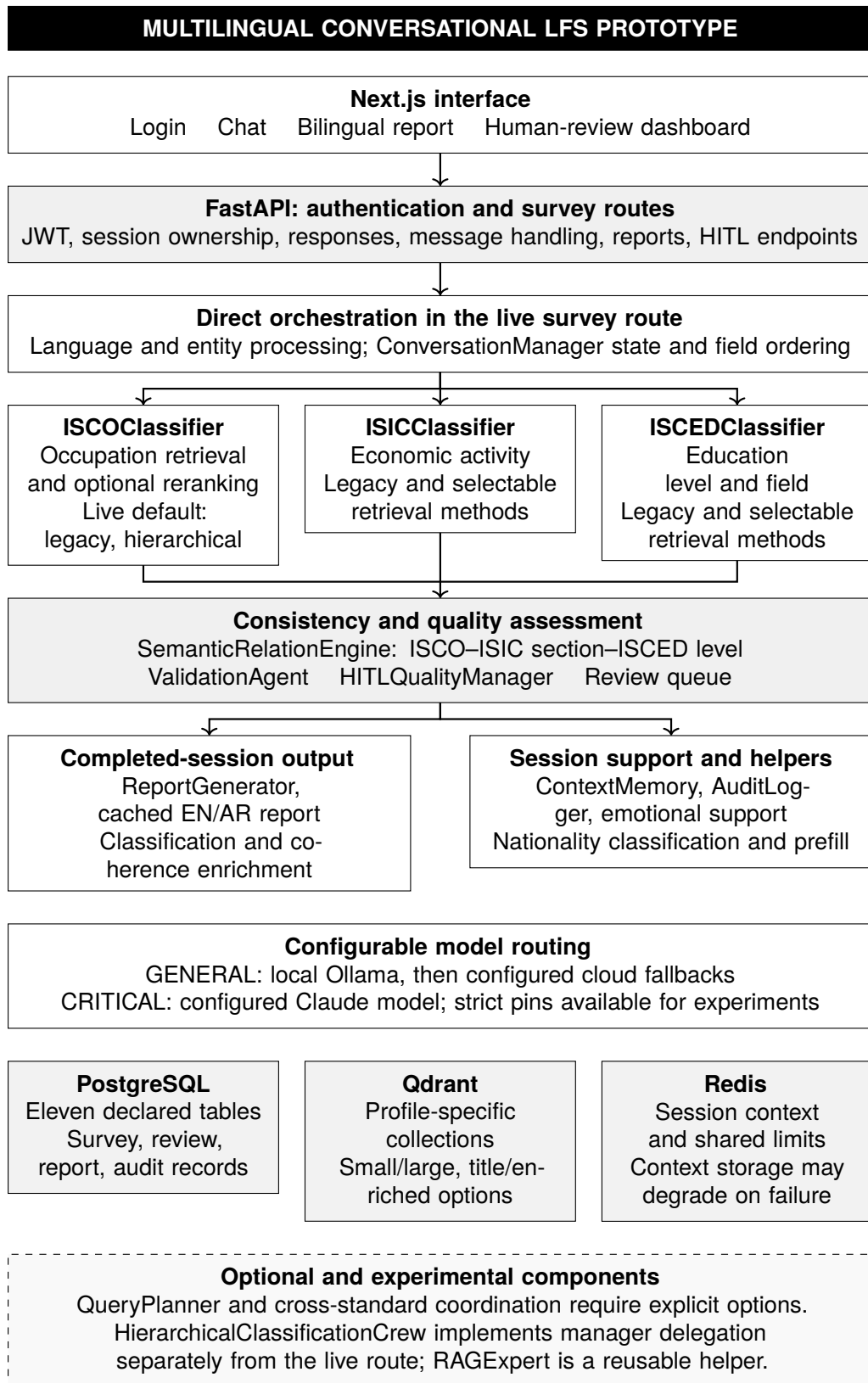


Figure 3.1: Implemented component boundaries and principal data paths. The live API uses direct orchestration; experimental CrewAI delegation is separate. Connections represent logical dependencies. Chapter 4 describes operational limitations.

ters clarification. Field order is recomputed when routing answers change. Required-field presence leads to a summary for confirmation or correction; confirmation advances to completion.

The branches contain 56 distinct implemented fields. A field is not necessarily one question or one turn: clarification adds turns, prefill can remove requests, and optional fields are not all completion requirements. Demographic and feedback fields are shared across employed, unemployed, and outside-labour-force paths. Employment-specific fields cover occupation, duties, industry, hours, work arrangements, benefits, and work quality. Conditional fields include field of study for higher attainment, wage/contract items for paid employees, secondary-job hours, active job-search methods, previous-employment details, platform-work details, and an Emiratization item.

These rules adapt the project questionnaire; they do not demonstrate full conformity with ILO labour-force measurement [9]. Initial status is extracted from a respondent’s label, and the implemented unemployed-to-outside-labour-force reassignment fires when both search and availability are negative. A field instrument must establish reference-period work, temporary absence, availability, search, and applicable exceptions through appropriate questions.

Rules R01–R10 check required fields, hours parseability/range/outliers, employment-status contradictions, and full-time/part-time hour mismatches. The wage gate and labour-force reassignment are conversation-control rules rather than part of this validation set. The current API reports validation findings but does not universally block confirmation when data are invalid. Repeated clarification failure can accept an unnormalised answer. Independent questionnaire and routing validation therefore remain necessary.

3.5 ISCO-08 Retrieval Pipelines

Catalogue profiles. The official-source catalogue contains 436 unit groups, 130 minor groups, 43 sub-major groups, and 10 major groups. Armed Forces unit groups use internal padded forms 0110, 0210, and 0310; this is a code-normalisation convention. Legacy and official profiles are distinct. The live API constructs a bare `ISCOClassifier()`, whose default is `legacy`, rather than the best-performing official enriched E5-large configuration reported in Chapter 6.

Hierarchical retrieval. Search traverses major, sub-major, minor, and unit levels using parent filtering and beam search. The legacy hierarchical score combines stage similarities as

$$c = 0.10s_1 + 0.20s_2 + 0.20s_3 + 0.50s_4. \quad (3.1)$$

This is a heuristic confidence, not a calibrated probability. Official-profile search settings are recorded in their evaluation artifacts and summarised in Appendix F.

Flat retrieval. Direct search uses one unit-group collection; ancestry can be recon-

structed from the taxonomy. Official-text enrichment adds definitions, tasks, and included occupations to embedding text. E5-small uses 384-dimensional vectors and E5-large 1024-dimensional vectors [19]. Query and indexing configuration must match the chosen profile.

Reranking and review. An optional LLM chooses among retrieved candidates. A similarity score at least 0.92 permits bypassing reranking, and confidence below 0.70 is a basic review trigger. Experimental configurations can enable additional ambiguity policies. These thresholds are not universal risk bounds. Legacy keyword/synonym handling differs from official-definition enrichment. Collection creation does not specify custom HNSW tuning parameters.

Chapter 6 compares the recorded retrieval configurations. Those offline results do not measure the default interview pipeline or an unseen deployment population.

3.6 ISIC Rev. 4 Industry Classifier

ISIC Rev. 4 organises economic activities into section, division, group, and class levels [18]. The keyword table has 134 entries: 121 match official classes and 13 are non-standard. The enriched flat index excludes the latter, covering 121 of 419 official classes. This is partial coverage.

The default API uses the legacy keyword/LLM pipeline; hierarchical and flat retrieval are alternative configurations. Examples include 6201 (Computer programming activities) and 4100 (Construction of buildings). Successful examples do not establish population accuracy. Occupation and employer industry are separate variables: an ICT professional may work outside the ICT industry.

3.7 ISCED Classifier

ISCED 2011 attainment (levels 0–8) [16] is distinct from ISCED-F 2013 field of education [17]. Attainment uses rules; field classification supports keyword and retrieval configurations. Legacy combined confidence is

$$c_{\text{education}} = 0.4c_{\text{level}} + 0.6c_{\text{field}}. \quad (3.2)$$

The score is an implementation heuristic. National qualifications and ambiguous descriptions require independent mapping validation.

The field table contains 63 entries, of which 61 match official definitions. The enriched flat index excludes the two non-standard entries. Attainment accuracy cannot be inferred from field-classification accuracy, and coverage must refer to the actual index evaluated.

3.8 Semantic Relation Engine

The Semantic Relation Engine checks an ISCO occupation against an ISIC section and an ISCED 2011 attainment level. Detailed ISCED-F field codes are not inputs to its core score. Compatibility tables encode project-authored occupational expectations; an official correspondence source for these exact rules has not been established.

Let q_I and q_E represent industry and attainment compatibility. The initial score is

$$r = \begin{cases} 0.55q_I + 0.45q_E, & \text{both signals available,} \\ q_I, & \text{industry only,} \\ q_E, & \text{attainment only,} \\ 0.80, & \text{neither available.} \end{cases} \quad (3.3)$$

With m moderate violations, the final score is

$$S = \text{round}_4(\min\{1, \max\{0, r + 0.15m(1 - r)\}\}). \quad (3.4)$$

The coherent flag uses $S \geq 0.70$. The missing-signal default of 0.80 is an assumption, not observed evidence of compatibility.

Industry restrictions can produce HIGH violations. Attainment severity depends on distance from an expected interval: one level is LOW, two MODERATE, and more than two HIGH. Stricter minima are defined for sub-major groups 22, 23, 25, and 26. These expectations can flag valid non-traditional careers; they are not mandatory occupational entry requirements.

The engine exposes a separate caller-confidence adjustment: +0.10 at $S \geq 0.90$, +0.05 at $S \geq 0.70$, -0.05 at $S \geq 0.50$, and -0.20 otherwise. The live API scores coherence and queues HIGH violations, but does not generally apply this adjustment to returned/persisted ISCO confidence. An optional coordinator can reconsider an uncertain occupation using cross-standard evidence. These are distinct mechanisms, not one calibrated fusion model.

The rule-validation and endpoint tests in Chapter 6 check specified rules and queue creation. They do not establish sensitivity, specificity, or fairness against independently adjudicated respondent data.

3.9 Technology Stack

3.10 End-to-End Data Flow

A turn performs authenticated session lookup, context retrieval, language processing when enabled, state update, relevant classifications, optional coordination, coherence scoring, op-

Table 3.2: *Main technologies; Python versions are pinned in `requirements.txt` and evaluation profiles are separately configured.*

Layer	Technology
Frontend	Next.js 14.2.35, React 18
Backend	FastAPI 0.131.0, SQLAlchemy 2.0.46, Alembic 1.18.4
Agent/task framework	CrewAI 1.9.3
Retrieval	Qdrant client 1.17.0; multilingual E5 profiles
Persistence	PostgreSQL 15; Redis client 7.2.0
Model access	Ollama and configured cloud-provider adapters
Deployment specification	Docker Compose; outstanding schema/startup issues are documented in Chapter 4

tional engagement analysis, validation at the validation state, database commit, context persistence, and logging. Completion persists collected fields and may trigger fallback classification, person-register update, and quality review. Reports aggregate stored responses and can run further enrichment/narrative generation.

Call count and latency vary with state, input, fast mode, and model configuration. Retrieval latency and end-to-end interview duration measure different workloads. The review also found that refined classifications or corrected titles could differ from stored responses. These defects were repaired and covered by regression tests; passing tests constrain the behaviours they exercise and do not establish complete end-to-end correctness.

Chapter 4

Implementation

The prototype combines a Next.js interface, a FastAPI backend, PostgreSQL records, Qdrant retrieval, Redis context storage, and configurable language models. This chapter describes its components and current limitations. Evaluation configurations are distinguished from interactive defaults.

4.1 Development Environment

Table 4.1 lists selected dependencies pinned in `requirements.txt`. Direct package pins do not by themselves guarantee identical transitive dependencies, model downloads, or services across installations.

Table 4.1: *Selected dependencies pinned in the application manifest.*

Library	Version	Purpose
crewai	1.9.3	Agent and task abstractions
fastapi	0.131.0	REST API server
sqlalchemy	2.0.46	Relational persistence
alembic	1.18.4	Database migrations
qdrant-client	1.17.0	Vector database client
sentence-transformers	5.2.3	Multilingual embeddings
redis	7.2.0	Context and authentication-limit state
python-jose	3.5.0	JWT signing and verification
pyotp	2.9.0	One-time-code generation
sendgrid	6.12.5	Email delivery adapter
twilio	9.10.2	SMS delivery adapter
langdetect	1.0.9	Probabilistic language detection
rank-bm25	0.2.2	Sparse retrieval baseline
pytest	9.0.2	Automated tests

Gulf-Arabic normalisation uses the hand-built `_GULF_NORMALISE` dictionary in `language_processor.py`. It performs lexical substitutions rather than morphological analysis through an external dialect toolkit. Embedding model selection is independent of the Python library version.

4.2 Deployment Configuration

The repository provides a Windows script and Docker Compose configuration:

```
start.bat
docker compose -f docker/docker-compose.yml up --build
```

The Windows script installs dependencies, runs `alembic upgrade head`, and starts the native application with containerised infrastructure. Compose defines PostgreSQL, Redis, Qdrant, backend, and frontend services. Ollama is configured separately through its server URL. Table 4.2 lists configured local endpoints.

Table 4.2: *Default local service configuration.*

Service	Local endpoint	Role
Next.js	<code>http://localhost:3000</code>	Login, chat, report, review
FastAPI	<code>http://localhost:8000</code>	Authentication and survey API
PostgreSQL 15	Port 5432	Survey, review, and audit records
Qdrant	Port 6333	Profile-specific retrieval collections
Redis 7	Port 6379	Context and rate-limit persistence
Ollama	Port 11434	Separately configured local model service

The October 2026 review found a first-boot defect in the backend Dockerfile. Its initialization command imported the declarative `Base` and invoked `create_all()` without first importing model declarations, so the metadata was empty and a fresh PostgreSQL volume received no tables; a previously initialized database concealed the problem, and the image omitted the root Alembic configuration. The image now packages that configuration and runs `alembic upgrade head` before starting the server, verified against a fresh PostgreSQL 15 volume with a repeated upgrade and a full table and column comparison. The procedure is validated for that environment, not for an institutional deployment target.

The `/ready` endpoint checks database connectivity with `SELECT 1`, not schema presence, classifier availability, or end-to-end survey readiness. Qdrant collection counts and installed model counts depend on the profiles built and models downloaded; they are not architectural constants.

4.3 Relational Schema and Persistence

Eleven SQLAlchemy tables are declared in `backend/database/models.py`.

Table 4.3: *Declared relational tables and responsibilities.*

Table	Purpose
<code>users</code>	Identifiers, email, phone, and account status
<code>otp_codes</code>	Codes, expiry times, and consumption flags
<code>survey_sessions</code>	Ownership, language, status, and timestamps
<code>survey_responses</code>	Field answers and occupation classifications
<code>audit_logs</code>	Recorded interaction events
<code>data_access_logs</code>	Explicitly recorded data-access events
<code>agent_decision_logs</code>	Decision summaries and confidence values
<code>quality_reviews</code>	Session quality scores and review records
<code>hitl_queue</code>	Classification and coherence escalations
<code>person_register</code>	Attributes retained for later prefill
<code>survey_report_records</code>	Cached bilingual reports and enrichment

Migrations introduce the initial schema, HITL tables, soft-delete timestamps, response revision links, and report enrichment columns. ISIC, ISCED, and semantic-coherence JSON are stored alongside report text, enabling those fields to survive cache reads.

The response model provides `supersedes_id` and `deleted_at` for revision history. The October 2026 review found write paths that bypassed them: duties-based occupation refinement could be returned to the interface without replacing the stored classification, and a corrected job title could remain only in context when entity extraction created no new response, so completion skipped the stored title and the report kept the earlier answer. Refinements and corrections now save active revisions, retire superseded review entries, and reclassify even when extraction is skipped; completion and reports read the active revision.

Deleting a session sets a timestamp and retains its records. The prefill query previously included deleted sessions and inactive revisions, so deleted answers could enter a new interview; it now excludes both. Soft deletion implements retention and visibility checks in selected routes, not verified erasure.

4.4 Classification Catalogues and Embeddings

Official ISCO-08 catalogue profiles contain 436 unit groups [8]. Their loader verifies counts and source hashes. Enriched profiles add definitions, tasks, and example occupations from the source workbook to title-only embedding text. Small and large multilingual E5 encoders use separate profiles; results are presented in Chapter 6.

The live survey route initializes `ISCOClassifier()` without an explicit profile. The constructor defaults to the legacy catalogue, hierarchical retrieval, and the small embedding model. This differs from the official, enriched, large-encoder flat configuration used for the best full-heldout retrieval result. Neither official-profile coverage nor that result should be attributed to the live default.

Retrieval may be followed by LLM reranking when an agent is available and the score is below 0.92. The unpinned constructor uses `TaskType.GENERAL`; an explicitly specified reranker uses the strict model factory. Failure to initialize the unpinned agent permits retrieval-only classification. The threshold is a configured rule, not evidence of a calibrated probability of correctness.

ISIC and ISCED classifiers expose legacy and retrieval-based methods. Education level and field-of-study code are separate outputs. Each experiment and the interactive route must be identified by their selected methods, rather than assumed to use every available retrieval option.

4.5 Language Processing

Five languages are exposed: English, Arabic, Urdu, Hindi, and Tagalog, with separate `ar` and `ar-gulf` codes for Arabic variants. An `other` code handles uncertain or unsupported input. Detection combines `langdetect.detect_langs()` with script heuristics, including a Devanagari fast path for Hindi and Gulf lexical markers. A `GENERAL` model agent extracts named entities; failure can produce an empty entity list.

Code-switch detection identifies Arabic–Latin or Devanagari–Latin input when both script proportions meet the 0.10 threshold. It cannot establish every language switch, particularly between languages sharing a script. Normalisation is similarly limited to the dictionary entries.

Translated prompts and options are implemented, but complete multilingual routing has not been established. Hindi and Urdu employment-status buttons send translated labels that the status extractor does not normalize reliably. Clarification fallback can store an invalid status and truncate the subsequent field order. The operational validation is English only. Interface availability must therefore be distinguished from validated interview routing, comprehension, and completion.

4.6 Validation and Quality Review

The `ValidationAgent` checks ten deterministic rule identifiers:

- R01 requires employment status and industry; employed respondents additionally require job title, weekly hours, and employment type.

- R02 requires supplied hours to be numeric.
- R03 checks weekly hours in $[1, 168]$; R04 warns above 80.
- R05 and R06 flag an unemployed respondent's current job title or positive weekly hours.
- R07 flags a current job title outside the labour force.
- R08 warns about full-time work below 20 hours; R09 warns about part-time work at 35 hours or above.
- R10 flags zero hours for an employed respondent.

These are selected consistency checks, not a complete implementation of international labour-force classification requirements. R01 requires industry for every status, although inactive questionnaire paths do not collect current industry; those requirements need reconciliation.

When deterministic checks contain no errors, `validate()` attempts a CRITICAL semantic-model check. The live route calls this method at the validation state without explicitly disabling that stage. Initialization or runtime failures may be caught by the route. A validation result is therefore not guaranteed on every turn, and validation is not a mandatory completion gate.

The Semantic Relation Engine applies hand-built occupation–industry and occupation–education plausibility rules. It consumes an ISCED 2011 level, not an ISCED-F detailed field. HIGH violations can create review entries during a turn; report generation adds a backstop for later detection. This is an implemented escalation path under tested conditions, not empirical validation of the crosswalk. Appendix C describes its rule-consistency evidence.

4.7 Frontend

Four pages provide OTP login, conversation, bilingual report display, and supervisor review. Chat renders the next field, progress, classifications, and a structured correction picker. Reports display occupation, industry, education, and coherence summaries.

The skipped-question display uses a fixed 155-question comparator without a verified mapping to the implemented field order. The *AI Verified* badge depends only on occupation confidence and can overlook semantic escalation. The supervisor table reads a confidence property differing from `ai_confidence` in the API. These displays are not evidence of measured efficiency or completed review.

4.8 Security and Data Governance

JWT authentication and ownership checks protect ordinary session routes. Email-code request limits and shared OTP attempt counters use Redis, with an in-process fallback. CORS uses a configured origin allowlist, and middleware adds response security headers.

HITL endpoints do not enforce a supervisor role or authority over the target respondent. An ordinary account can list the global queue and change another respondent's classification. SMS-code requests lack request limits. These gaps prevent a claim of readiness to collect participants' personal data.

Email and phone identifiers remain linked to survey records. The person register is a prefill store, not a pseudonymisation mechanism. Logging and retention-purge helpers do not establish comprehensive access logging, scheduled purging, or regulatory compliance. GENERAL routing can use configured cloud providers when the local model is unavailable, so local inference is not a guaranteed boundary for respondent text. The proposed study needs an approved data-management plan addressing these issues (Appendix D).

4.9 Testing and Scope of Verification

Tests under `backend/tests/` and `eval/` exercise rules, routing, retrieval contracts, persistence, and evaluation calculations. Many substitute model or service responses. Passing tests establish checked software properties, not multilingual accuracy, regulatory compliance, or complete service reliability.

The documented 16 September 2026 run recorded 2,567 passing tests and one slow test deselected. Historical runs with several `PYTHONHASHSEED` values demonstrate repeatability under tested conditions, not absence of every defect or nondeterministic behavior. A focused review on 3 October 2026 passed 194 existing conversation, orchestration, and evaluation tests while identifying 28 defects. Those defects were repaired and covered by added regression tests; a full suite run on 7 October 2026 recorded 2,341 passing tests and one slow test deselected. A participant study remains outside this evidence.

The legacy BM25 comparison uses developer-authored synthetic cases. The principal occupation benchmark and its limitations are described in Chapter 5.

Chapter 5

Evaluation Methodology

The evaluation separates occupation coding on an external reference dataset, industry and field-of-study coding on generated text, software consistency tests, and a proposed participant study. These evidence types answer different questions. None establishes the prototype’s accuracy or usability in a representative Labour Force Survey population.

5.1 ISCO-08 Evaluation: WISCO

WISCO provides externally published occupation titles and ISCO codes [15]. The project restructures its English, Arabic, Urdu, Hindi, and Tagalog title variants into benchmark records. Missing occupation–language combinations are skipped. These are reference dictionary entries supplied by the dataset, not interviews with survey respondents and not translations generated by this project’s benchmark builder.

The prepared data contain 4,232 source occupation records and 20,760 occupation–language records. Each variant retains its source occupation identifier and published unit-group label. Development and heldout partitions are constructed by this project; they are not official WISCO train/test partitions.

5.1.1 Grouping, Splits, and Leakage Checks

The initial per-occupation split kept language variants of each source record together, but an audit found exact duplicate titles associated with different source identifiers across the two partitions. The second builder uses union-find groups: source records sharing a normalized title within a language are merged, and each group receives a deterministic assignment based on a fixed-seed hash. This produces 2,013 development records and 18,747 heldout records.

An additive third version partitions the development pool into 1,371 development and 642 validation records, retaining exactly the same 18,747 heldout record identifiers. The development/heldout assignment uses seed 42; the second development/validation decision uses seed 4224. The build summary reports no source-family or exact normalized-title overlap across

the resulting partitions. This check does not rule out all semantic similarity, pretrained-model exposure, or dependence between language variants. Near-duplicate candidates were reported separately rather than automatically removed.

Development records support exploration, validation records support configuration selection, and heldout records support confirmation. Multiple reported experiments nevertheless reuse the same heldout partition. Their comparisons therefore require attention to repeated testing and possible adaptive selection; a fixed partition alone does not establish that every experimental decision was made without reference to earlier heldout results.

5.1.2 Configurations and Paired Comparisons

Flat retrieval searches one unit-group collection. Hierarchical retrieval traverses major, sub-major, minor, and unit-group collections with parent filtering and beam selection. Comparisons use matching benchmark record identifiers and published labels. Catalogue enrichment and encoder size are varied separately, then jointly, to examine their contributions to retrieval performance. Results and configuration-specific reranking settings are given in Chapter 6.

The official title-only, enriched, small-encoder, and large-encoder profiles differ from the legacy profile selected by the live survey route. An offline score applies to its recorded configuration, not to every configuration that shares the classifier class.

Reranking checks use explicitly configured language models where recorded, including cloud providers and local Ollama models. Provider selection, whether reranking actually fired, and fallback behavior must be reported alongside accuracy. A requested model name is not evidence that its inference completed successfully.

5.2 ISIC Rev. 4 and ISCED-F 2013: Synthetic Evaluation

WISCO does not supply the industry or field-of-study labels required by these evaluations. No accessible dataset meeting this project’s requirements for labelled multilingual verbatim responses was obtained. The recorded IPUMS correspondence states that the requested original response text was unavailable to that provider. This establishes a limitation of the project’s data access, not the nonexistence of all labelled ISIC or ISCED datasets.

Synthetic cases are generated from official ISIC Rev. 4 descriptions and ISCED-F 2013 descriptions [17, 18]. A classification definition and language instruction are supplied to an LLM, which produces a short respondent-style paraphrase. The generating class is retained as the intended gold label.

The label is fixed by the prompt; it is not independently adjudicated against the generated sentence. Generation can introduce ambiguity or change the meaning, even when the output passes automatic checks. Moreover, generation and retrieval both use taxonomy descriptions. Performance consequently measures recovery of an intended generating category under this

construction, not accuracy on independently collected respondent text. These cases are suitable for regression checks and controlled comparisons, subject to that limitation.

Generation used a Groq API and was accumulated across sessions under provider token limits. Recorded model and prompt identifiers, source settings, and their provenance limits appear in Appendix F. The target combines matched catalogue classes with six language codes, including the Gulf-Arabic variant. Actual coverage and its distribution are reported with each result; the target size is not presented as completed coverage.

5.2.1 Quality Checks and Historical Contamination

The separate quality checker examines script contamination, predominant script, verbatim English-title leakage, and refusal patterns without calling an LLM. These checks assess usability of the text and adherence to generation instructions. They do not verify every language’s fluency, dialect authenticity, or semantic agreement with its intended label.

Two refusal responses were included in the original 449-row dataset used for the reported comparison. They were detected after that evaluation. The generator was subsequently updated to reject refusal patterns, and affected data were repaired. The original measured result remains a result on the earlier contaminated dataset; it is not reclassified as a clean evaluation. A repaired or expanded dataset requires a fresh completed run before a new accuracy claim can be made. The recorded rerun limitation and the older result’s scope are described in Chapter 6.

The Gulf-Arabic subset also supports a comparison with and without lexical normalisation. It contains generated dialect-oriented text rather than a native-speaker-validated sample of real dialectal interviews. The normaliser’s result is therefore specific to that sample and rule set.

5.3 Secondary Synthetic-Corpus Comparison

The earlier harness in `eval/legacy_thesis_ch6/evaluate.py` compares BM25, flat dense retrieval, and hierarchical retrieval. Its `TEST_CASES` list contains 100 English descriptions with developer-authored ISCO labels; `ARABIC_TEST_CASES` provides a separate 30-case Arabic set. The script does not document independent professional double-coding or adjudication, and it contains no equivalent Hindi, Urdu, or Tagalog case sets.

This corpus is a secondary engineering evaluation. Its label provenance and language coverage differ from WISCO. The BM25 class is also reused by the main harness, but sharing an implementation does not make the two datasets equivalent.

5.4 Metrics and Artifact Integrity

Occupation top-1 accuracy is the proportion of eligible cases whose predicted code equals the published gold code. The analyzer also reports agreement at major, sub-major, and minor prefixes. Industry and field-of-study metrics distinguish their respective hierarchy levels; coarse-level agreement is not interchangeable with detailed-code accuracy. Empty predictions from a completed inference attempt are failures under exact-match scoring.

Pre-reranking top-3 candidate coverage is a retrieval metric:

$$\text{coverage@3} = \frac{\#\{\text{eligible cases with gold in the first three candidates}\}}{\#\{\text{eligible retrieval cases}\}}.$$

It differs from accuracy of the final reranked prediction. The `analyze.py` implementation previously excluded blank `gold_rank_in_pool` values, although the harness leaves this value blank when retrieval misses the gold code, which inflated top-3 coverage. A missing gold candidate now counts as a failure wherever a valid pool exists, and unavailable telemetry is identified separately. Corrected summaries were written alongside, not over, the original raw predictions: full-scale retrieval top-3 is $61/130 = 46.92\%$, previously reported as $61/108 = 56.48\%$. Top-1 figures were checked separately and are unchanged.

The exact-match analyzer previously did not enforce `evaluation_status` and could label a dry-run row as measured. It now distinguishes unmeasured rows explicitly, and accuracy evidence is restricted to verified inference runs identified by run records and statuses.

For non-BM25 runs, the CSV encoder field and configuration hash currently use a fixed small-model name even for large-encoder profiles. Encoder identity must consequently be checked against the selected catalogue profile and run configuration. This provenance defect requires metadata correction, but does not alone establish that the underlying predictions were generated by the wrong encoder.

5.5 Planned Participant Pilot

The proposed parallel-arm pilot has not been executed. The repository’s ethics submission log records no submitted application or completed participant recruitment. The draft protocol proposes 30 participants, 15 assigned to AI administration and 15 to a traditional interviewer. Randomisation details, recruitment criteria, translated consent materials, and the data-management plan remain to be finalized through the institutional review process.

Planned outcomes are completion time, item nonresponse, internal contradiction rate, respondent satisfaction, and per-interview cost. This sample is intended for feasibility assessment, not a definitively powered effectiveness comparison. A separate 30-session synthetic, English-only operational run was completed (Section 6.4.2); it supplies no participant satisfaction or randomized-arm evidence.

Any actual pilot must use equivalent questionnaires, document withdrawals, record the installed configuration, and define how contradictions and costs are measured. Before deployment with participants, the known authorization, routing, persistence, and data-governance gaps described in Chapter 4 must be addressed. Proposed participant information appears in Appendix D.

5.6 Statistical Methods and Limits

The evaluation code computes Wilson 95% intervals for proportions and the exact two-sided McNemar test for paired correctness outcomes. For s successes among n cases, with $\hat{p} = s/n$ and $z = 1.96$, the implemented Wilson limits are

$$\frac{\hat{p} + z^2/(2n) \pm z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}.$$

McNemar’s test uses the discordant pairs: b cases correct only under method A and c cases correct only under method B. The implementation evaluates the two tails of a binomial distribution with $b + c$ trials and null probability $1/2$. These functions are implemented directly in Python rather than through SciPy’s statistical API.

The reported aggregate calculations treat occupation–language records as separate cases. Variants of a shared occupation identifier are related, and the current calculations are not adjusted for that clustering. The resulting intervals and p -values describe the case-level analysis under its assumptions; they are not population uncertainty estimates for real LFS respondents. A confirmatory extension would use occupation-group-aware uncertainty estimates and a declared plan for multiple comparisons. Statistical significance does not remove dataset, catalogue, or workflow limitations.

For the participant pilot, the draft proposes Welch’s t -test for completion time, Mann–Whitney U for satisfaction, and Fisher’s exact test for binary outcome comparisons, with corresponding effect sizes. These are planned analyses, not results. The unit of analysis for item-level outcomes and dependence of answers within an interview must be specified before data collection; participants, not individual answers, are the randomisation units.

Chapter 6

Results and Discussion

This chapter addresses the four research questions through occupation-coding benchmarks, synthetic multi-standard experiments, rule-based validation, and operational tests. These evidence sources measure different properties: WISCO supports a controlled comparison of occupation-title coding; generated texts assess taxonomy recovery; scripted interviews assess software operation. None validates respondent experience or population estimates from a labour force survey.

6.1 ISCO-08 Retrieval Design (RQ1)

6.1.1 Benchmark and Statistical Interpretation

The principal comparison uses the 18,747 records assigned to the frozen WISCO v2 heldout partition, derived from an externally sourced multilingual occupation-title dataset [15]. Exact-match accuracy counts predicted four-digit codes equal to their reference code, divided by all evaluated records. The official catalogue has 436 unit groups and is distinct from the application’s legacy profile. No LLM reranker was used in the full-partition comparisons below.

Queries use the E5 `query`: prefix, unit-normalised embeddings, and an embedding batch size of one. Flat search requests five candidates from its profile’s unit-group collection. The canonical hierarchical run uses beam width two, description-based first-stage retrieval, and five leaf candidates per explored final branch. Keyword anchoring supplies the first stage on 4,100 records; the remaining 14,647 use semantic first-stage retrieval. A strict guard rejects flat fallback and incomplete stage evidence, with a 30-second stage budget. The encoder dimensions are 384 for e5-small and 1,024 for e5-large.

Tables report counts, percentage-point differences, and 95% Wilson score intervals [21]. Paired comparisons use the two-sided exact binomial form of McNemar’s test [12]: if b records are correct only under the first configuration and c only under the second, the null distribution is $\text{Binomial}(b + c, 1/2)$. Records are matched by case identifier and reference

label. Extremely small probabilities are computed without floating-point underflow and are not reported as zero.

The intervals and tests are nominal record-level summaries. Language variants and related occupation titles need not be independent; no cluster-adjusted uncertainty analysis was performed. Moreover, the same partition was reused for later comparisons after earlier outcomes had been examined. Later configurations therefore provide retrospective benchmark results, rather than an independent confirmatory test of a configuration selected without access to its results. Group-aware splitting reduces development–test overlap but cannot remove selection bias caused by repeated inspection of test performance [1].

6.1.2 Flat and Hierarchical Retrieval

Table 6.1: *Exact four-digit ISCO-08 accuracy on the frozen WISCO partition ($n = 18,747$), official catalogue and e5-small embeddings.*

Configuration	Correct	Accuracy	95% Wilson CI
Flat retrieval	3,973	21.19%	[20.61%, 21.78%]
Hierarchical retrieval	1,941	10.35%	[9.93%, 10.80%]

Flat retrieval exceeds the tested hierarchical configuration by 10.84 percentage points. Paired outcomes comprise 1,341 records correct under both systems, 2,632 correct only under flat retrieval, 600 correct only under hierarchical retrieval, and 14,174 incorrect under both. The nominal exact comparison gives $p = 1.86 \times 10^{-301}$.

This result does not support the hypothesis that the implemented hierarchical filter improves exact coding. Error propagation is a plausible explanation: an incorrect ancestor restricts later retrieval to an unsuitable branch, whereas flat retrieval preserves competition across unit groups. The finding concerns the particular catalogue, embeddings, and search policies tested here; it does not rank every flat and hierarchical method.

6.1.3 Embedding Model and Catalogue Enrichment

The title-only representation contains a code and short occupation title. Enrichment adds definitions, tasks, and example occupations from the official ILO source [8]. The encoder changes from `multilingual-e5-small` to `multilingual-e5-large`; the reference codes are retained across these four flat configurations.

The paired discordant counts (b, c) relative to the baseline are (941, 2535), (743, 2872), and (668, 4371), respectively. The highest observed accuracy is 40.95%, with nominal 95% interval [40.24%, 41.65%]. It exceeds enriched e5-small by 8.40 percentage points, with $(b, c) = (1125, 2699)$ and $p = 5.40 \times 10^{-147}$. Combining enrichment and encoder size

Table 6.2: Flat retrieval configurations on the same 18,747 records. Differences and nominal exact McNemar probabilities are relative to the title-only e5-small baseline.

Configuration	Correct	Accuracy	Δ (pp)	p
Title-only, e5-small	3,973	21.19%	—	—
Title-only, e5-large	5,567	29.70%	+8.50	7.34×10^{-167}
Enriched, e5-small	6,102	32.55%	+11.36	8.59×10^{-293}
Enriched, e5-large	7,676	40.95%	+19.75	7.69×10^{-663}

improves on either change alone. It does not demonstrate statistical independence between their mechanisms; no factorial interaction analysis was specified.

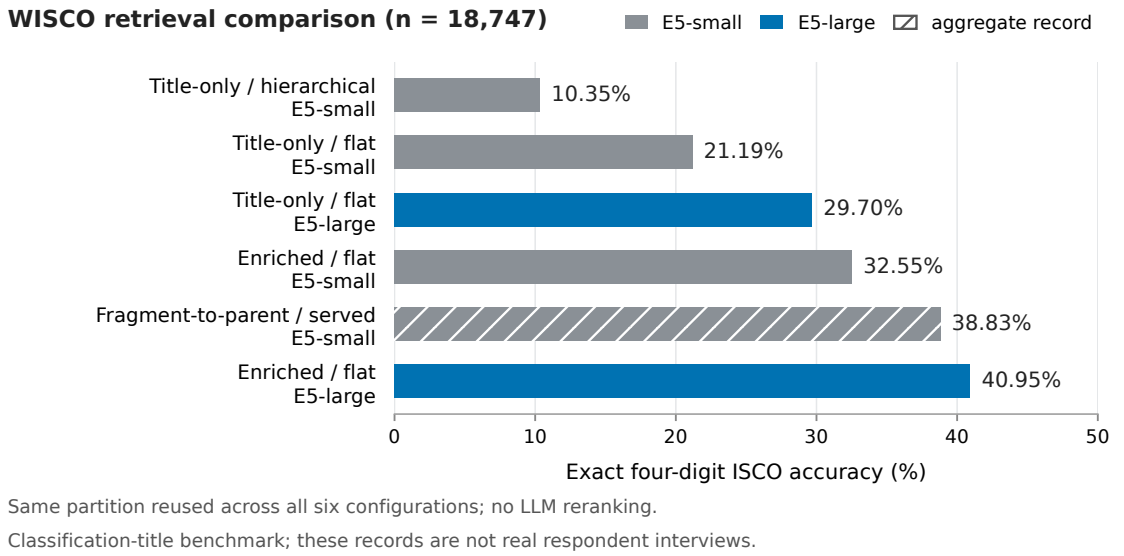


Figure 6.1: Exact ISCO-08 accuracy across six recorded configurations on the same 18,747 WISCO records. Later configurations reuse the benchmark; the figure summarises observed results rather than an independent confirmatory comparison. The hatched bar is the configuration the application serves (Section 6.1.4); its count comes from a preserved aggregate audit record rather than released per-case predictions, so it is verified for internal consistency rather than for case-identity against the other five. Settings and provenance limits appear in Appendix F.

With e5-small fixed, enrichment changes which codes dominate retrieval. Assignments to code 6122 decrease from 468 to 24, and assignments to 5165 from 276 to 31. This is consistent with short catalogue entries attracting unrelated queries. A frequency closer to the reference distribution does not, by itself, establish the correctness of individual predictions.

Every language benefits from the combined configuration, although substantial error remains. The larger encoder alone does not improve English in this run (37.98% to 37.59%); combined gains should not be attributed to encoder size alone.

Table 6.3: *Per-language exact-match accuracy: title-only e5-small versus enriched e5-large. Differences are computed before rounding.*

Language	n	Baseline	Enriched large	Δ (pp)
Arabic	3,762	14.62%	37.77%	+23.15
English	3,818	37.98%	56.91%	+18.94
Hindi	3,793	23.07%	40.89%	+17.82
Tagalog	3,766	14.47%	33.17%	+18.69
Urdu	3,608	15.33%	35.53%	+20.21

6.1.4 Fragment-to-Parent Retrieval in the Served Configuration

The configurations above are offline comparisons. The local application serves a different design, adopted because the enriched e5-large index could not be loaded reliably on the available machine (Section 6.6). Each official catalogue entry is divided into title, definition, and included-example fragments. Fragments are indexed individually and retrieve their parent unit group; a parent’s score is the maximum over its own fragments, with a fixed weight of 0.5 on fragment evidence. All 436 unit groups remain eligible for every query.

This is distinct from the parent-filtered hierarchical traversal of Table 6.1, despite the shared word: no level decision restricts the candidate set, and no code is removed by an earlier stage. The encoder is `multilingual-e5-small` over the same enriched official catalogue, so the comparison below holds encoder and catalogue fixed and changes only the retrieval unit.

Table 6.4: *Served fragment-to-parent configuration against its same-encoder dense baseline, on the same 18,747 records. The third row is the offline configuration of Table 6.2, whose executed encoder is not established.*

Configuration	Correct	Accuracy	Status
Enriched dense, e5-small	6,102	32.55%	Same-encoder baseline
Fragment-to-parent, e5-small	7,279	38.83%	Served locally
Enriched flat, intended e5-large	7,676	40.95%	Encoder unresolved

Fragment-to-parent retrieval has a nominal 95% interval of [38.13%, 39.53%]. Against the same-encoder dense baseline the paired discordant counts are $(b, c) = (720, 1897)$, a gain of 1,177 codes (+6.28 percentage points, nominal exact McNemar $p = 3.75 \times 10^{-121}$). Against the offline intended-large output it is 397 codes lower (−2.12 points, $(b, c) = (2328, 1931)$, $p = 1.26 \times 10^{-9}$). The latter difference is real in the saved predictions, but because that run’s executed encoder is not established (Section 6.6) it does not isolate an encoder cause.

Every language improves over the same-encoder baseline, with the smallest gain in Arabic.

Table 6.5: *Per-language exact-match accuracy of the served fragment-to-parent configuration against its same-encoder dense baseline. Differences are computed before rounding.*

Language	n	Dense small	Fragment-to-parent	Δ (pp)
Arabic	3,762	25.15%	27.94%	+2.79
English	3,818	54.85%	63.91%	+9.06
Hindi	3,793	33.80%	40.18%	+6.38
Tagalog	3,766	23.13%	29.85%	+6.72
Urdu	3,608	25.19%	31.60%	+6.40

The configuration is bound to a recorded serving identity: catalogue, fragment map, encoder revision, encoder weight digest, and aggregation parameters are hashed in the preserved audit record (Appendix F). Unlike the configurations in Table 6.2, its per-case predictions are not published with this thesis, so the figures here are recomputed from that recorded summary rather than from released per-case outputs. A later catalogue-alignment variant, which projects the small-encoder query into the stored large-encoder parent space through a ridge map fitted only on paired official catalogue vectors, reached 39.16% on this partition (+0.34 points, 63 codes). It was not adopted. A promotion gate declared before that run required a positive gain, case-level $p < 0.05$, and no per-language regression; Tagalog top-1 accuracy fell, so the gate failed on its third condition. The variant was also inert on the two smaller splits (development +1 case, $p = 1.0$; validation +3 cases, $p = 0.66$), and it depends on stored vectors whose encoder provenance is unresolved. It is therefore recorded as a non-adopted research candidate rather than a result. All figures use the same reused partition and remain offline reference measurements, not Labour Force Survey field validation.

Because the figures above are re-derived from a preserved summary rather than released per-case records, the live application was also exercised directly to confirm that it serves this method and not another. The running API’s diagnostic occupation route reported the fragment-to-parent method identifier for every query, and returned the expected unit groups for a short spread of unambiguous titles — software engineer to 2512, nurse to 2221, taxi driver to 8322, primary school teacher to 2341, accountant to 2411, and the Arabic rendering of “software engineer” to 2514 — at 126–851 ms per query. **This is a configuration check on six easy titles, not an accuracy measurement:** it establishes that the deployed path is the configuration reported here, and says nothing about accuracy beyond the 38.83% figure.

6.1.5 LLM Reranking and Corrective Retrieval

The available paired checks show small or absent changes. On 500 development records, e5-large retrieval and its Groq-reranked variant both classify 146 correctly (29.20%); one code changes without changing correctness. On 642 validation records, title-only e5-small increases from 117 to 120 correct (18.22% to 18.69%, $b = 0$, $c = 3$, $p = 0.25$). Four codes

change, all for the same occupation, “Air force captain”. Enriched e5-small increases from 209 to 211 correct (32.55% to 32.87%, $b = 0$, $c = 2$, $p = 0.50$).

A local-model corrective-retrieval check, which reformulates the query before another retrieval attempt, returns 31/60 correct with and without the mechanism (51.67%). Nine final codes change, including two gains and two losses in correctness ($b = 2$, $c = 2$, $p = 1.00$). These experiments do not demonstrate a material benefit for the tested settings. They do not establish equivalence or show that all LLM interventions are ineffective. The samples overlap the development process, use different configurations, and are not independent replications. Failure to reject a null hypothesis is not proof of a zero effect [20].

6.1.6 Zero-Shot Coding and Query Translation

An exploratory zero-shot run asks `gpt-oss-120b` for a code without a candidate list. Its saved output contains 324 records, with 61 correct (18.83%). Restricting the official flat e5-small baseline to the same identifiers gives 86/324 correct (26.54%). There are 31 zero-shot-only and 56 retrieval-only successes, yielding nominal exact $p = 0.00967$. Retrieval performs better on this recorded subset; the quota-limited partial run is not a random representative sample or a full comparison of prompting strategies.

A translation experiment uses 60 development records, 15 in each of four non-English languages. Local `qwen2.5:3b` translates the query into English before retrieval against the English-sourced enriched catalogue. The experiment tests alignment with the reference text; it does not assume that translation always improves multilingual retrieval.

Table 6.6: *Exploratory query translation: exact-match successes out of 15 records per language.*

Language	Original query	English translation
Arabic	6/15 (40.0%)	9/15 (60.0%)
Hindi	6/15 (40.0%)	10/15 (66.7%)
Urdu	4/15 (26.7%)	5/15 (33.3%)
Tagalog	4/15 (26.7%)	8/15 (53.3%)
Overall	20/60 (33.3%)	32/60 (53.3%)

Thirteen records change from incorrect to correct and one changes in the opposite direction ($p = 0.00183$). This 20.0-point improvement motivated the further testing reported below. The full-partition translated evaluation did not complete; that result remains exploratory. Translation is an optional classifier setting and is not enabled in the live API’s default configuration.

That further test was then carried out against the served fragment-to-parent configuration (Section 6.1.4), on a larger sample and a different retrieval method. Its design was recorded

before the run: development cases only, 400 records stratified as 100 per non-English language with a fixed seed, both arms scoring the identical records with the identical classifier so that only the text reaching the encoder differs, and no production change on any outcome.

Table 6.7: *Query translation against the served configuration, 400 development records. b and c are the paired discordant counts (original-only and translation-only correct); p is the nominal exact two-sided McNemar probability.*

Language	n	Original	Translated	Δ	b / c	p
Arabic	100	32	43	+11	12 / 23	0.089
Hindi	100	37	36	−1	15 / 14	1.000
Tagalog	100	23	21	−2	12 / 10	0.832
Urdu	100	34	28	−6	11 / 5	0.210
Overall	400	126 (31.50%)	128 (32.00%)	+0.50 pp	50 / 52	0.921

The 20.0-point exploratory result did not replicate. In aggregate the translated arm is indistinguishable from the original (+0.50 percentage points, $p = 0.921$), so no translation gain is claimed for this configuration and the setting remains disabled.

The per-language columns, however, show why the aggregate is flat: Arabic gains 11 records while Urdu loses 6 and Hindi and Tagalog are unchanged within noise, so the languages cancel. Arabic is the only arm approaching conventional significance ($p = 0.089$ on 35 discordant pairs) and it moves in the direction the earlier sample predicted, but it does not reach it and is not reported as an improvement. The methodological point is that an intervention can be selective by language and still average to zero; reporting only a pooled figure would have concealed the entire effect. The original arm’s 31.50% is consistent with this configuration’s recorded per-language accuracy for these four languages, so the sample is not anomalous — it excludes English, which is this configuration’s strongest language. Per-case outputs are retained locally but not published, since they embed benchmark text and gold labels.

6.1.7 Error Structure

Exact four-digit accuracy reports how often a configuration is completely right and says nothing about the shape of the remaining error. Because ISCO-08 is hierarchical, a prediction can miss the unit group while still identifying the correct occupational family, and that distinction matters for a coding assistant: a near miss is a candidate for human adjudication, whereas a prediction in an unrelated major group is a failure to recognise the occupation at all. Table 6.8 recomputes both enriched configurations at every level of the classification from their preserved per-case predictions.

The gap between the first and last rows is the substantive finding. The better configuration places 63.59% of records in the correct major group while coding only 40.95% to the exact

Table 6.8: Accuracy at each ISCO-08 hierarchy level on the same 18,747 records. Each row is the same prediction evaluated against a shorter prefix of the gold code.

Level	Enriched, e5-small		Enriched, intended e5-large	
	Correct	Accuracy	Correct	Accuracy
Major group (1 digit)	10,399	55.47%	11,922	63.59%
Sub-major group (2 digit)	8,319	44.38%	10,068	53.70%
Minor group (3 digit)	7,143	38.10%	8,919	47.58%
Unit group (4 digit)	6,102	32.55%	7,676	40.95%

unit group, so a substantial part of what the exact-match figure counts as failure is a correct reading of the occupation at a coarser granularity. Expressed over the 11,071 unit-group failures directly: 4,246 of them (38.4%) are in the correct major group, and 1,243 (11.2%) reach the correct minor group and miss only the final digit. The weaker configuration shows the same structure less strongly (34.0% of its failures in the correct major group), so enrichment and the intended larger encoder improve the classification at every level rather than only at the exact-match threshold. No record received an empty prediction in either configuration.

The residual error is structured rather than diffuse. The single most-missed gold code is 2310 (university and higher education teachers) at 229 records, 2.1% of all failures, and its most frequent substitutions are other teaching and professional codes. The most frequent single confusion is gold 6112 predicted as 9211 on 59 records, which pairs a skilled agricultural occupation with an elementary agricultural labouring code — a distinction that rests on the respondent’s role rather than the subject matter of the work, and which short title text may not carry at all. Such pairs suggest that part of the remaining error reflects genuine underdetermination in a title-only benchmark rather than retrieval failure, which is a limitation of the evaluation material as much as of the method.

The confidence score does not separate correct from incorrect predictions. Mean confidence is 0.808 on correct records and 0.786 on incorrect ones for the better configuration, and 0.829 against 0.802 for the weaker one. A gap of roughly two hundredths cannot support a confidence-based routing threshold, which is the empirical content of the statement elsewhere in this thesis that these scores are uncalibrated heuristics and that human review is required for every classification. Establishing a usable review threshold would require calibration against labelled decisions, which has not been done.

Per language, the near-miss share of failures is nearly constant across the four non-English languages at 36.7–37.7%, while English reaches 44.4%. English therefore leads not only on exact accuracy (56.91% against 33.17–40.89%) but also in how close its failures fall, which is consistent with the English-sourced catalogue text discussed in Section 6.1.6.

These figures are recomputed by `eval/analyze_isco_error_structure.py` from the same preserved predictions used elsewhere in this chapter; it loads no model and

selects no configuration.

6.2 Multi-Standard Classification and Coherence (RQ2)

6.2.1 Synthetic ISIC and ISCED-F Benchmark

The benchmark contains 307 ISIC class descriptions and 142 ISCED-F field descriptions generated from taxonomy-grounded prompts in six language varieties. Each reference label is the generating category, not independent expert coding of respondent text. The comparison measures industry or education-field category recovery, not ISCED attainment.

Table 6.9: *Synthetic category recovery: legacy keyword/LLM pipeline versus enriched flat e5-large retrieval.*

Standard	n	Legacy correct	Flat correct
ISIC Rev. 4	307	38 (12.38%)	245 (79.80%)
ISCED-F 2013	142	21 (14.79%)	127 (89.44%)
Combined	449	59 (13.14%)	372 (82.85%)

Combined nominal Wilson intervals are [10.33%, 16.58%] and [79.09%, 86.05%]. There are 9 legacy-only and 322 flat-only successes ($p = 5.54 \times 10^{-83}$). The aggregate weights standards by sample size; it is not an equally weighted task average. Flat accuracy ranges from 78.08% in Tagalog to 88.46% in English on these generated descriptions.

The result is conditional on restricted catalogue coverage. Legacy tables contain 134 ISIC and 63 ISCED-F entries, including 13 and 2 entries without an official-source match. Enriched flat indexes exclude these and contain 121 officially matched ISIC classes and 61 ISCED-F detailed fields, short of complete coverage [17, 18].

Two of the 449 descriptions were later identified as LLM refusal responses. The table retains the original evaluated data and was not recomputed after replacing those rows. Generation was subsequently expanded to 1,091 records and refusal filtering revised, but no completed accuracy evaluation of that expanded corpus is available. Shared taxonomy definitions between generation and retrieval may make category recovery easier than coding spontaneous answers; independent expert-labelled data are necessary.

6.2.2 Semantic Relation Engine

The SRE was checked against 61 constructed cases: 20 expected coherent, 12 LOW, 14 MODERATE, and 15 HIGH. All returned severities agree with the expected rule outcomes. Three endpoint runs repeat those same cases: all 15 HIGH cases escalate, and none of the 46 other

cases escalates through this severity-based trigger. The repeated executions demonstrate repeatability of the rule and queue integration, not three independent validation samples.

Expected outcomes are constructed from the project’s compatibility rules. Agreement supports implementation correctness, not sensitivity or specificity against independently adjudicated contradictions. The score is a heuristic compatibility index, not a calibrated probability of correctness. Hand-built occupation–industry and occupation–education constraints require expert assessment, particularly for atypical but legitimate employment and educational histories.

6.2.3 Coordination and Query Planning

A separate 60-case English synthetic benchmark supplies paired occupation, industry, and education inputs absent from WISCO. Deterministic coordination reconsiders uncertain occupation codes using cross-standard compatibility. Query planning decomposes descriptions into up to two subqueries and reconciles their retrieval results.

Table 6.10: *Occupation accuracy on 60 synthetic combined cases. Discordant counts refer to baseline-only and intervention-only successes.*

Intervention	Baseline	Intervention	(b, c)	p
Coordination, legacy	16/60	16/60	(0, 0)	1.00
Coordination, enriched large	29/60	29/60	(0, 0)	1.00
Query planning, legacy	16/60	17/60	(1, 2)	1.00

Coordination leaves all occupation predictions unchanged in both configurations. Query planning changes 21 occupation codes and adds one correct prediction overall (1.67 percentage points); industry and education predictions are unchanged. These small experiments do not establish equivalence, adequate power, or a general limit on coordination. An unchanged output may reflect conservative triggers or insufficient alternatives, rather than successful ambiguity resolution. A subsequently implemented manager-mediated delegation path has no corresponding comparative result in this evidence set.

6.3 Multilingual and Dialect Behaviour (RQ3)

Table 6.3 supports multilingual occupation-title retrieval, with substantial language differences. It does not evaluate mixed-language interviews, entity extraction, translation fidelity, or dialect detection against independently annotated conversational data.

6.3.1 Gulf Arabic Normalisation

The recorded experiment uses 61 generated Gulf Arabic descriptions. The marker dictionary identifies 8/61 (13.11%), and unnormalised retrieval classifies 49/61 correctly (80.33%). The A/B summary reports the same accuracy after normalisation with no discordant correctness outcomes. Per-case normalised predictions were not retained with the evaluated CSV, limiting independent reproduction of the paired comparison.

Table 6.11: *Recorded exploratory normalisation results ($n = 61$). The normalised arm is an experiment summary, not a saved per-case artifact.*

Measure	Recorded result
Marker detections	8/61 (13.11%)
Unnormalised classification successes	49/61 (80.33%)
Normalised classification successes	49/61 (80.33%)

No improvement was observed. Generated dialect labels are not independently adjudicated ground truth, and low marker coverage may mean that little input text is changed. This experiment cannot establish general dialect robustness or show that normalisation is unnecessary in other tasks.

6.4 Interview Feasibility and Computational Behaviour (RQ4)

6.4.1 Local Computational Measurements

The enriched-large manifest records Windows 10, Python 3.11.9, an Intel Family 6 Model 154 Stepping 4 processor with 12 logical processors, and 7.71 GiB RAM; CUDA is reported unavailable. It records `sentence-transformers` 5.2.3, `PyTorch` 2.10.0, `transformers` 4.46.3, and `qdrant-client` 1.17.0. The canonical baseline artifacts lack a comparably complete execution hardware manifest. The three rows therefore describe their recorded runs; differences across run families are not an isolated comparison under verified identical hardware and software conditions. Appendix F records the saved configuration details and their provenance limits.

The enriched-large run trades higher accuracy for increased compute and memory demand. Its highest sampled process RSS is 1,822.91 MiB; mean RSS across recorded case boundaries is 1,243.31 MiB. These snapshots do not capture allocation peaks that subside before a case finishes. A separate e5-small microbenchmark reports 19.08 ms mean embedding time over 120 calls, excluding import and loading. The Qdrant memory snapshot is 94,928,896 bytes (90.53 MiB) across the database process, not a per-index measurement.

Table 6.12: *Local retrieval-only latency. End-to-end timing covers occupation classification in the evaluator, excluding initialisation and conversational interaction. Runs are sequential rather than concurrent.*

Configuration	Mean (ms)	Median (ms)	95th percentile (ms)
Title-only flat, e5-small	31.11	30.08	42.70
Title-only hierarchical, e5-small	133.86	137.38	182.61
Enriched flat, e5-large	208.12	194.63	291.13

The load-test artifact records 42/42 successful scripted request sequences. Final states are collecting, clarifying, or validating rather than completed interviews. The sequence-based success criterion does not establish a capacity of 42 simultaneously completed interviews. These measurements concern local development hardware; production throughput, deployment cost, and Claude-backed latency remain unmeasured.

6.4.2 Synthetic Operational Validation

The approval-dependent respondent pilot in Section 5.5 was not conducted. No institutional ethics submission, recruited participants, or pilot responses are recorded. A separate single-arm test drives 30 synthetic English profiles through the HTTP API. Occupation, industry, and education inputs reuse 30 cases from the combined benchmark; other fields use schema-valid scripted answers. This tests local conversation flow and completion, not human reading, response generation, or satisfaction.

Table 6.13: *Synthetic operational results through the local API ($n = 30$, English).*

Measure	Observed result
Sessions completed	30/30 (100%)
Mean scripted processing time	111.64 s
Mean turns per session	45.43
Sessions with at least one clarification turn	30/30 (100%)
Sessions with a HIGH-severity SRE signal	18/30 (60.0%)
Sessions marked incoherent by the SRE	22/30 (73.3%)
Exact-match occupation codes assigned	5/30 (16.67%)

The live API uses the legacy hierarchical occupation profile and legacy industry and education pipelines. It does not use the official flat configuration achieving 21.19% or the enriched-large configuration achieving 40.95% on WISCO. Thus 5/30 cannot confirm either WISCO figure. Conversation extraction and cached-result behaviour also differ from classification of isolated benchmark inputs.

Timing excludes human reading, typing, and decision time; report generation is outside the driver’s recorded completion interval. Clarification frequency may reflect canonical answers as well as conversation logic. SRE flags concern the system’s predictions, not independently verified contradictions in respondent answers: classification errors can create apparent incompatibility. Completion supports feasibility of the tested flows, not burden reduction, satisfaction, comparative cost-effectiveness, or field reliability. An isolated enriched-large evaluation of the same 30 cases did not complete under insufficient memory; its WISCO result is not substituted for this unmeasured operational-profile result.

6.5 Research Questions and Objective Status

Table 6.14: *Evidence against the research questions and linked objectives.*

Question / objectives	Evidence and remaining scope
RQ1 / O2	Tested flat retrieval exceeds the hierarchical configuration; the enriched configuration with the intended larger encoder reaches 40.95% on the reused WISCO partition, with that run’s executed encoder unresolved. The served fragment-to-parent configuration reaches 38.83% with a recorded serving identity. Independent confirmation and improved accuracy remain necessary.
RQ2 / O3–O5	Multi-standard coding and severity-triggered escalation are implemented. Synthetic recovery and constructed rule fixtures support restricted claims; independent coding and crosswalk validation remain absent.
RQ3 / O6	Retrieval is evaluated in five languages. Gulf normalisation has an exploratory generated-data result; code-switching and dialect behaviour require independent annotation and evaluation.
RQ4 / O1, O7, O8	Conversation, consistency rules, reports, and review flows have implementation tests and scripted evidence. Respondent burden, satisfaction, and comparative field performance remain unmeasured.

6.6 Threats to Validity

Construct validity. Exact occupation match, generated taxonomy recovery, rule-fixture agreement, and scripted completion are different outcomes. None directly measures labour-force-estimate quality, human coding agreement, or respondent experience. SRE compatibility is heuristic plausibility, not factual verification or a calibrated error probability.

Data-capture validity. Answer extraction is opportunistic: a message is scanned for several fields, not only the one just asked. This reduces the number of questions a respondent must answer, but it also means a field can be populated from an incidental keyword in an answer to a different question. A direct check against the running system illustrated both halves. Replying to the education question with an occupation sentence set the industry field from the word

“software” alone, and stored the sentence itself as the education answer, which the education classifier then coded to an unrelated level and field. The state machine correctly rejected the answer and repeated the question, and a valid reply overwrote both the stored value and its classification, so a completed interview is not left with the spurious entry. The residual exposure is an interview abandoned at that point, where the retained value and its derived code would be the rejected text. No respondent data is affected, because no interviews with participants have been conducted; this is a measurement-design property to resolve before any are, and it is of the same kind as the numeric-answer restriction already applied to wage capture.

Internal and statistical validity. Repeated test-partition inspection limits confirmation of selected configurations. Related multilingual records and taxonomy-derived texts introduce dependence unaccounted for in nominal intervals and tests. Exploratory comparisons lack a prespecified adjustment for multiple testing. Small nonsignificant experiments do not establish equivalence or absence of practically relevant effects; no study-specific power analysis or equivalence margin was specified.

External validity. WISCO is an occupation reference rather than data collected in the conversational LFS. Synthetic generation shares definitions with retrieval, restricts coverage, and lacks independent expert adjudication. Its results do not establish performance on spontaneous, incomplete, ambiguous, or atypical descriptions. Operational validation is English-only, with no human participants or traditional-interview comparison.

Reproducibility and deployment validity. Preserved CSVs, JSON summaries, and manifests support recomputation, but metadata must be interpreted with the executed configuration. The original enriched-large CSV names e5-small in its encoder metadata although the profile selects e5-large; profile mappings and build/run records document the intended encoder. The CSV and later manifest also contain different commit identifiers, and the CSV’s recorded commit does not contain the complete enriched-large profile implementation. The executed working-tree state is therefore incompletely recorded; profile labels alone cannot establish an exact source reconstruction. The original artifacts are retained. Normalised Gulf predictions and several planned expanded evaluations lack completed per-case outputs. Local memory constraints prevented expanded evaluations and caused instability during an attempted API switch to e5-large. The best offline configuration therefore cannot be used as the served system’s performance estimate; the served configuration is measured separately in Section 6.1.4, on the same reused partition and subject to the same reuse limitation. Its own per-case predictions are not published, so its figures rest on a hashed summary rather than released records.

Chapter 7

Conclusion and Future Work

7.1 Summary

This thesis develops a multilingual conversational system for collecting labour force questionnaire responses, coding occupation, industry, educational attainment, and field of education, and routing uncertain or incompatible results to human review. State-based conversation control, retrieval and keyword classifiers, a heuristic Semantic Relation Engine, and persistent reports and audit records support the implementation. The evaluation distinguishes controlled coding benchmarks from synthetic operational evidence.

For RQ1, flat ISCO-08 retrieval exceeds the tested hierarchical configuration on 18,747 WISCO records: 21.19% versus 10.35% exact-match accuracy. Catalogue enrichment, combined with the intended larger encoder, raises the highest observed accuracy to 40.95% (7,676 correct), although that run's executed encoder is not established from its preserved records. This identifies an engineering direction and shows that the tested hierarchical restriction is not beneficial in this setting. Later configurations were compared after results from the same partition were inspected, so the best result is a retrospective benchmark finding requiring independent confirmation. It is not the accuracy of the live API. The configuration actually served locally is fragment-to-parent retrieval, which reaches 38.83% (7,279 correct) on the same partition and improves on its same-encoder dense baseline by 6.28 percentage points (Section 6.1.4).

Reading only the exact-match figure understates what the retrieval achieves and overstates how usable it is, in two opposite directions. The same predictions place 63.59% of records in the correct major group while coding 40.95% to the exact unit group, and 38.4% of the unit-group failures sit inside the correct major group, so much of the measured error is a correct reading of the occupation at a coarser granularity rather than a failure to recognise it (Section 6.1.7). Against that, the confidence score separates correct from incorrect predictions by about two hundredths, so it cannot be used to decide which classifications need human attention. For an assisted-coding tool the practical consequence is that the system is more

useful than 40.95% suggests and less self-aware than any single accuracy figure implies: it frequently knows the occupational family, and it cannot currently tell the reviewer when it does not.

For RQ2, multi-standard coding and deterministic compatibility checks are implemented. On 449 taxonomy-grounded synthetic descriptions, enriched flat retrieval recovers 82.85% of generating categories compared with 13.14% for the legacy pipeline. The result concerns restricted coverage and includes two subsequently identified refusal descriptions. Agreement on 61 constructed SRE cases and repeatable escalation demonstrate rule and queue correctness, not independent contradiction validation or empirical score calibration.

For RQ3, occupation retrieval is evaluated in English, Arabic, Hindi, Tagalog, and Urdu. Enriched-large retrieval improves on the baseline in every language, with substantial remaining errors and language gaps. Translation has a positive 60-record development result; Gulf normalisation has no observed gain in the recorded synthetic experiment. Conversational code-switching and general dialect robustness remain unestablished.

For RQ4, 30 synthetic English profiles complete scripted interviews through the local API. This supports operational feasibility of the tested flows, not respondent burden, satisfaction, or comparative cost-effectiveness. The participant pilot was not conducted; no institutional ethics submission is recorded.

7.2 Contributions

C1 — Comparative retrieval evidence. Flat and hierarchical ISCO-08 retrieval and comparisons of catalogue richness and encoder size provide evidence for configuration choices. The observed advantage of the tested flat configuration is reported without claiming universal architectural superiority. Negative and exploratory findings accompany the highest observed accuracy.

C2 — Heuristic cross-standard review support. A deterministic SRE connects occupation, industry, and educational attainment through explicit compatibility rules and severity-triggered review. Its heuristic score and hand-built constraints are inspectable and tested; their substantive validity and calibration remain open.

C3 — Integrated multilingual prototype. A state-based questionnaire flow supports five languages, treats Gulf Arabic as an Arabic variety, and incorporates conditional questions, clarification, confirmation, and ten deterministic consistency rules. Multi-standard outputs, authentication, audit records, bilingual reports, and review mechanisms share a containerised service stack. Implementation support is distinguished from independently measured language quality and deployment readiness.

C4 — Evaluation and diagnostic artifacts. Automated tests, controlled benchmark exports, synthetic operational experiments, and preserved prediction artifacts support inspection and recomputation. Method labels and configuration records help distinguish retrieval results

from fallback behaviour. These artifacts identify measurement gaps as well as observed performance; they do not themselves establish respondent benefit.

7.3 Limitations

L1 — No participant validation. Burden, satisfaction, accessibility, and comparative interview outcomes remain unmeasured. Scripted synthetic sessions cannot resolve them.

L1b — Uncalibrated confidence. Retrieval similarity does not distinguish correct from incorrect predictions to any useful degree: mean confidence differs by roughly two hundredths between them (Section 6.1.7). Every classification therefore carries a human-review requirement, and no selective-review policy, workload estimate, or automation rate can be derived from these scores until they are calibrated against labelled decisions.

L2 — Reused benchmark and dependent observations. Later configurations were informed by results on the same WISCO partition. Nominal intervals and paired tests do not account for related titles or multilingual variants, and exploratory comparisons lack multiplicity adjustment. The selected configuration requires a fresh test set.

L3 — Restricted coverage and synthetic labels. Enriched flat indexes contain 121 officially matched ISIC classes and 61 ISCED-F fields, rather than complete coverage. Generating categories are not expert labels for spontaneous descriptions. Two refusal rows remain in the reported 449-record result; expanded-corpus evaluation is incomplete.

L4 — Heuristic cross-standard knowledge. An authoritative correspondence mapping was not established in this project. Constructed expected outcomes do not estimate the SRE's error rate on respondents. Atypical work and education may be legitimate, so violations require review rather than automatic correction.

L5 — Incomplete language and intervention evaluation. Language results primarily measure occupation-title retrieval. Conversational code-switching, entity extraction, dialect detection, and translation lack comprehensive independent annotation. Small intervention checks do not establish equivalence or a general absence of benefit.

L6 — Offline and deployed configurations differ. The API retains legacy coding pipelines whose accuracy cannot be inferred from official-profile offline results. The Claude-backed critical tier has not been demonstrated end-to-end in the recorded evaluation, and an API switch to the larger encoder encountered local memory instability.

L7 — Limited runtime evidence. Recorded local measurements exclude human interaction and do not establish production availability, throughput, or cost. Some paired experimental outputs and expanded evaluations remain incomplete.

L8 — Application correctness and access control. The routing, answer-extraction, persistence, reviewer-authorization, measurement, and fresh-installation defects found in the October 2026 review were repaired and covered by regression tests (Chapter 4). The evidence for those repairs is automated testing and local verification, not operation with participants, so it

constrains the behaviours exercised rather than establishing deployment readiness.

7.4 Future Research Directions

F1 — Approval-dependent respondent study. Obtain ethics approval before recruitment, then conduct the planned pilot with a documented comparison condition. Prespecify outcomes and analysis; measure language accessibility, satisfaction, and time including human interaction. Use feasibility results to design an adequately powered subsequent study rather than assume a small pilot answers all effectiveness questions.

F2 — Independent coding evaluation. Obtain permitted occupation, industry, and education text with independent expert labels, adjudication, and inter-coder reliability. Separate partitions by source occupation or respondent and freeze the selected configuration before final testing. Use cluster-aware uncertainty where records share a source title or multilingual reference.

F3 — Catalogue and reference governance. Expand officially matched ISIC and ISCED-F coverage, maintain source versions and provenance, and validate exclusions and code normalisation. Complete quality-controlled expanded synthetic evaluation while separately pursuing independent respondent validation.

F4 — Expert validation and calibration of review rules. Assess compatibility with domain experts and independently labelled combinations. Measure false-positive review burden and missed contradictions, including legitimate atypical cases. Calibrate scores or candidate gaps on separate data and select review thresholds from validated error and workload trade-offs; raw cosine similarity is not a probability. The measured two-hundredth separation between correct and incorrect predictions sets the baseline any such calibration must beat, and suggests that a usable review signal will have to come from evidence the current score does not encode — candidate-gap structure, cross-standard agreement, or the hierarchy level at which the prediction stabilises — rather than from rescaling the similarity itself.

F4b — Coarse-grained coding as a reportable output. Because major-group accuracy exceeds exact-code accuracy by more than twenty percentage points, evaluate whether a deliberately coarser assignment, with the unit group offered only when supported, is more useful to a statistical office than a single exact code of uncertain quality. This requires agreement from the statistical authority on what granularity is acceptable for which estimates, and measurement of coder effort under each option.

F5 — Controlled intervention studies. Evaluate translation, reranking, corrective retrieval, and coordination under matched inputs and pinned settings. Prespecify practical effects and, for equivalence questions, a margin and sample-size justification. Report failure rates, tokens, latency, and expert assessment of changed outputs. Measure candidate coverage over all eligible records, including those whose reference code is absent from retrieval.

F6 — Deployment and configuration validation. Test the selected profile on hardware

with sufficient memory, then assess API integration, concurrency, failures, and recovery before changing the default. Preserve configuration identities between offline and deployed results. Benchmark alternative critical-tier models on their actual tasks, rather than infer suitability from occupation-reranking outcomes. Re-verify the repaired routing, persistence, access-control, evaluation, and deployment behaviours in the intended deployment environment before participant use, since the present evidence is local and automated.

F7 — Other survey instruments. Investigate generalisation of state-based collection, multi-standard coding, and auditable review after establishing their validity, operational constraints, and respondent effects in the original LFS context.

Appendix A

LFS Questionnaire Fields

Table A.1 documents selected fields and canonical values from `ConversationManager`'s correction schema and dynamic field order. The item codes are implementation labels, not evidence that the prototype reproduces a complete official UAE questionnaire. Interview prompts vary by language and state. The schema describes intended values, not proof that every translated answer is normalized correctly.

Table A.1: *Selected implemented fields, canonical values, and routing gates.*

Code	Field	Canonical values or input	Routing
C1	employment_status	employed, unemployed, not_in_labour_force	All paths
C3	employment_nature	paid_employee, employer, self_employed, contributing_family_member	Employed
C5	job_title	Free-text current occupation	Employed; ISCO-08 input
C5a	job_duties	Free-text duties	Employed; occupation-refinement input
C6	industry	Free-text economic activity	Employed; ISIC Rev. 4 input
D7	contract_type	permanent, temporary, no_contract	Employed paid employees
E1	monthly_wage_range	AED bands: under_5000, 5000_to_10000, 10000_to_20000, 20000_to_30000, above_30000	Employed paid employees
F1	job_search_active	yes, no	Unemployed

Continued on the next page

Table A.1 — continued

Code	Field	Canonical values or input	Routing
F3	available_for_work	yes, no; availability within two weeks	Unemployed; F1=no and F3=no changes status to outside the labour force
B6	field_of_study	Free-text main field	All paths with education bachelor, master, or phd

Dynamic order also depends on nationality, secondary employment, platform work, and previous employment (Section 3.4). The wage gate belongs to the conversation manager, not the ValidationAgent’s rules. A known extraction defect can populate wages from an unrelated numeric answer and skip the later question. Hindi and Urdu status options also have a normalization defect. The table records intended routing rather than an assertion that every path is correct.

Appendix B

Agent Configuration Reference

Agents are constructed in Python. These listings are descriptive configuration summaries, not executable YAML files. Interactive defaults are separated from optional evaluation features.

B.1 — Occupation Classifier

Listing B.1: *Current occupation-classifier configuration summary.*

```
interactive_route: ISCOClassifier()
catalogue_profile: legacy default
retrieval: hierarchical default; flat available by explicit option
encoder: multilingual-e5-small default
official_profiles:
  unit_groups: 436
  encoders: multilingual-e5-small or multilingual-e5-large
  text: title-only or enriched definitions/tasks/examples
hitl_threshold: 0.70
skip_rerank_threshold: 0.92
reranking:
  default_factory: TaskType.GENERAL
  explicit_model: get_llm_strict(model, temperature)
  enable_llm_false: retrieval only
  unavailable_default_agent: retrieval-only degradation
```

The official enriched large-encoder flat profile supplies the principal full-heldout retrieval result, not the live interview default. GENERAL routing prefers a configured local Ollama model and can fall back through cloud providers. CRITICAL routing selects the configured Claude model without the GENERAL fallback chain. The current occupation constructor itself selects GENERAL when unpinned, despite its older module documentation.

B.2 — Semantic Relation Engine

Listing B.2: *Implemented coherence rules and score settings.*

```
source: project-authored plausibility tables
inputs: ISCO-08 code, ISIC section, ISCED 2011 level
not_an_input: ISCED-F 2013 detailed field
raw_score:
  both_dimensions: 0.55 * industry_ok + 0.45 * education_ok
  one_dimension: available dimension's compatibility score
  neither_dimension: 0.80
partial_credit: 0.15 * moderate_count * (1 - raw_score)
score: clipped to [0, 1], rounded to 4 decimal places
coherent_threshold: 0.70
education_gap_severity:
  1: LOW
  2: MODERATE
  3_or_more: HIGH
confidence_adjustment:
  score_ge_0.90: +0.10
  score_ge_0.70: +0.05
  score_ge_0.50: -0.05
  otherwise: -0.20
education_submajor_minimums: {"22": 7, "23": 6, "25": 6, "26": 7}
industry_submajor_examples: {"22": ["Q"], "23": ["P"], "25": ["J"]}
```

Education severity uses the distance outside the allowed level range. Industry severity uses a different rule: an industry allowed at the major level but excluded by a stricter sub-major rule is MODERATE; an industry outside the applicable allowed lists is HIGH. Sub-major 11 has no special education override and inherits the managers' major-group range. Education rules concern expected attainment, not legally required qualifications or field-of-study correspondence.

Returning an adjustment value is distinct from applying it to a saved classification. These hand-set rules and thresholds do not produce an empirically calibrated probability or an official ILO/UNESCO crosswalk.

B.3 — Validation Agent

Listing B.3: *Validation rules and semantic stage.*

```
R01: employment_status and industry required for all statuses;
```

```
    job_title, hours_per_week, employment_type also required if
    employed
R02: supplied hours_per_week must be numeric
R03: hours_per_week must be within [1, 168]
R04: hours_per_week > 80 gives an outlier warning
R05: unemployed with current job_title gives a contradiction
R06: unemployed with hours_per_week > 0 gives a contradiction
R07: outside labour force with current job_title gives a
    contradiction
R08: full_time with hours_per_week < 20 gives a warning
R09: part_time with hours_per_week >= 35 gives a warning
R10: employed with hours_per_week = 0 gives an error
semantic_stage: attempted if deterministic checks contain no errors
semantic_model_factory: TaskType.CRITICAL
```

The route calls `validate()` without explicitly disabling its semantic stage, and catches validation failures. A successful result is not required for completion. Required fields also need reconciliation with inactive paths (Section 4.6).

Appendix C

Semantic Relation Engine: Validation Evidence

The available evidence checks implementation consistency and escalation. The 61-case fixture in `eval/sre_expanded_validation.py` assigns expected outcomes using the same project’s plausibility rules. Agreement on all 61 cases supports execution of those rules; it does not independently validate their accuracy on labour-force data.

The endpoint script reuses this fixture and checks for queue entries tagged as HIGH semantic incoherence. Three documented executions each recorded 15 of 15 expected HIGH cases queued and zero of 46 other cases queued for that reason:

Table C.1: *Repeated endpoint checks on the same 61-case fixture.*

Execution	Expected HIGH queued	Other cases with SRE HIGH entry
1	15/15	0/46
2	15/15	0/46
3	15/15	0/46

These are repeatability checks on the same cases, not independent respondent samples. The zero count concerns SRE HIGH entries, not every other cause of review. Queue creation does not establish reviewer authorization, review completion, or correctness of human decisions.

C.1 — Worked Score Example

For ISCO 2211, ISIC section Q, and ISCED level 7, both implemented checks pass. The professionals’ major-group education range is [6, 8], with a stricter minimum of 7 for sub-

major 22. Hence

$$r = 0.55(1) + 0.45(1) = 1, \quad m = 0, \quad s = \min\{1, r + 0.15m(1 - r)\} = 1.$$

The engine marks the combination coherent and returns an adjustment of +0.10. This calculation does not establish that every medical occupation requires a master's degree.

C.2 — Explanation Provenance

The implementation produces English and Arabic explanations. Its strong-alignment template currently attributes consistency to *ILO ISCO-ISIC-ISCED crosswalk tables*. That attribution does not match the project's hand-built tables and remains an unresolved product wording issue. The explanation should identify project-defined plausibility rules and their limits; the thesis does not endorse an official correspondence claim.

Appendix D

Participant Information and Study Protocol

This material describes a proposed study. The repository submission log records no submitted ethics application or completed recruitment. No participant results are reported. The draft remains subject to supervisor and institutional review.

D.1 — Draft Participant Information

Study title: Evaluation of a Multilingual Conversational AI System for Labour Force Survey Administration.

Principal investigator: Sivarama Krishnan Subbiah (2024MCS110007), M.Tech AI and Data Science, IIIT Kottayam.

Supervisor: Dr. Goutam Mali, IIIT Kottayam.

The proposed study compares a conversational AI interview with an interviewer-administered survey. Participation would involve employment and education questions followed by a satisfaction questionnaire. Intended languages are English, Arabic, Urdu, Hindi, and Tagalog; translated information and consent materials are not yet finalized.

Participation would be voluntary. Approved materials must state consent, withdrawal, retention, and deletion procedures before recruitment. Responses would be used for research rather than government, benefit, visa, or employment decisions. Automated codes may be inaccurate and are not official classifications of the participant.

The data-management design must distinguish contact identifiers from responses, define access, and explain whether text is sent to external model providers. The prototype links identifiers to responses and can use cloud fallback providers. Pseudonymisation is a proposed study requirement, not an implemented application guarantee. Logging tables do not replace consent or an approved retention plan.

D.2 — Readiness for the Proposed Study

Table D.1: *Documented status and remaining requirements for participant use.*

Requirement	Recorded status	Remaining work
Ethics application	Not submitted	Complete institutional forms and obtain review
Recruitment	Not started	Finalize approved eligibility and process
Information and consent	Draft	Finalize and verify translations
Pseudonymisation	Proposed	Separate identifiers and define access
Cloud processing	Configurable	Specify providers and participant disclosure
Supervisor authorization	Incomplete	Restrict review endpoints to authorized reviewers
Retention and withdrawal	Plan incomplete	Define durations, deletion, and purge operation
Multilingual workflow	Implemented with defects	Correct routing and validate supported languages
Pilot execution	Not started	Proceed after approval and readiness

The $n = 30$ parallel-arm proposal is a feasibility pilot, distinct from the English-only synthetic operational run in Section 6.4.2.

Appendix E

Project Structure

This tree lists principal source areas, not every file. Agent classes, deterministic helpers, and experimental orchestration are distinct.

Listing E.1: *Principal implementation and evaluation source areas.*

```
.
|-- requirements.txt
|-- requirements-dev.txt
|-- alembic.ini
|-- Dockerfile
|-- docker/docker-compose.yml
|-- start.bat
|-- backend/
|   |-- main.py
|   |-- api/
|   |   |-- auth_routes.py
|   |   |-- survey_routes.py
|   |-- auth/
|   |   |-- jwt_handler.py
|   |   |-- email_otp.py
|   |-- database/
|   |   |-- models.py
|   |   |-- connection.py
|   |   |-- migrations/
|   |-- agents/
|   |   |-- conversation_manager.py
|   |   |-- language_processor.py
|   |   |-- isco_classifier.py
|   |   |-- isic_classifier.py
|   |   |-- isced_classifier.py
|   |   |-- semantic_relation.py
```

```

|   |   |-- validation_agent.py
|   |   |-- hitl_quality_manager.py
|   |   |-- report_generator.py
|   |   |-- context_memory.py
|   |   |-- audit_logger.py
|   |   |-- emotional_intelligence.py
|   |   |-- nationality_classifier.py
|   |   |-- person_register.py
|   |   |-- cross_standard_coordinator.py
|   |   |-- query_planner.py
|   |   |-- hierarchical_classification_crew.py
|   |   |-- rag_expert.py
|   |-- llm/llm_client.py
|   |-- rag/
|   |   |-- hierarchical_store.py
|   |   |-- vector_store.py
|   |   |-- official_isco08_catalogue.py
|   |   |-- standard_hierarchical_store.py
|   |   |-- standard_flat_store.py
|   |-- tests/
|-- frontend/pages/
|   |-- index.js
|   |-- chat.js
|   |-- report.js
|   |-- supervisor_review.js
|-- eval/
    |-- run_eval.py
    |-- analyze.py
    |-- build_wisco_isco_benchmark_v2_group_split.py
    |-- build_wisco_isco_benchmark_v3_dev_validation_split.py
    |-- generate_synthetic_isic_iscedf_benchmark.py
    |-- validate_synthetic_benchmark_quality.py
    |-- sre_expanded_validation.py
    |-- sre_hitl_enforcement_endpoint_verification.py
    |-- legacy_thesis_ch6/

```

The live survey invokes components directly. `hierarchical_classification_crew.py` separately implements manager-led hierarchical CrewAI delegation for experimental use and is not imported by that route. Query planning is optional; RAGExpert is a reusable helper rather than live orchestration. Figure 3.1 distinguishes these paths.

Appendix F

Experimental Configuration and Artifact Provenance

This appendix identifies the preserved records supporting the principal occupation comparisons and synthetic generation. Table F.1 uses short run labels for cross-reference. All five occupation runs evaluate the same 18,747 heldout case identifiers. A profile identifies a catalogue and encoder configuration; it is distinct from a CSV configuration hash, an execution record, or a repository commit marker.

F.1 Occupation Retrieval Configurations

Table F.1: *Principal occupation-run configurations and their evidence sources. Encoder identities for B and D require profile/build evidence; the reduced or inconsistent CSV metadata alone is insufficient.*

Run	Profile and catalogue text	Encoder / dimensions	Search and record
A-flat	official_ilo2021_v1; title-only	E5-small / 384	Flat; full evaluator CSV
A-hier	official_ilo2021_v1; title-only	E5-small / 384	Hierarchical; full evaluator CSV
B	official_ilo2021_v1_e5large; title-only	E5-large / 1024	Flat; reduced CSV and collection-build manifest
C	official_ilo2021_v1_enriched; enriched	E5-small / 384	Flat; full CSV and later run manifest
D	official_ilo2021_v1_enriched_e5large; enriched	E5-large / 1024	Flat; full CSV and later run manifest
P	official_ilo2021_v1_enriched; enriched fragments	E5-small / 384	Fragment-to-parent; hashed audit summary

The encoders are `intfloat/multilingual-e5-small` and `intfloat`

/multilingual-e5-large. Enriched text adds official definitions and included occupations to the title representation. Each flat collection contains 436 unit groups. The A-flat/A-hier command record selects the official profile and disables LLM reranking; their CSVs record no reranker execution. The C/D manifests also explicitly record `use_llm_reranker=false`. B is described as retrieval-only in the archived experiment summary, but its reduced CSV does not independently record a reranker flag.

Run P is the configuration served by the local application and reported in Section 6.1.4. It differs from A–D in both retrieval unit and record type. Its index holds 2,833 fragment and parent vectors over the same 436 unit groups; a parent’s score is the maximum over its fragments, with fragment weight 0.5. Its preserved audit record binds the catalogue digest, fragment-map digest, selection digest, encoder identity `intfloat/multilingual-e5-small`, encoder revision `614241f6`, and an encoder weight digest, so the serving configuration is identifiable in a way runs B and D are not. Unlike A, C, and D, its per-case predictions are not released with this thesis; the reported counts come from that hashed summary. Its scores are an uncalibrated similarity blend and are not probabilities.

For A-flat/A-hier, the command record and source at the CSV-recorded commit `280ad76` establish the following settings. The evaluator requests five unit-group hits. A-flat returns the highest-similarity code and exposes up to three candidates. A-hier uses beam width two: up to two children are retained per parent at each non-leaf expansion, rather than a globally pruned two-path frontier. Its default stage-one mode is `description`, with a keyword major-group hint taking precedence when available. Recorded stage-one sources are 4,100 keyword hints and 14,647 semantic searches.

A-hier queries up to five unit-group hits per surviving minor-group branch and selects the path whose highest-ranked leaf has the greatest raw similarity. The weighted ancestor/leaf score supplies confidence, not the path-selection objective. With branch collapse disabled, leaf candidates are deduplicated across explored branches and ordered by their highest raw similarity; up to five are exposed as candidates. The strict hierarchical command requires genuine hierarchical execution and sets a 30,000 ms budget per stage, shared across that stage’s queries. The saved telemetry and eligibility analysis report no fallback or stage-budget failure in this completed run.

The historical A implementation prefixes queries with `query:`, uses normalized embeddings and query batch size one, and indexes `passage:`-prefixed catalogue text with cosine distance. The corresponding C source, identified by its recorded commit `3d10925`, likewise requests five flat hits and exposes three candidates. B’s reduced export does not preserve this complete parameter record. D’s later manifest identifies flat retrieval, its profile, and disabled reranking, but does not capture every search parameter. Missing historical settings should therefore not be filled from the interactive application’s current defaults.

F.2 Historical Records and Source Identity

The principal artifact locations are:

- **A-flat:** `eval/local_runs/wisco_official_tier1_precise_deadline_full_20260809T190821Z/flat/20260809T191133Z_wisco_official_tier1_precise_deadline_full_flat.csv`.
- **A-hier:** `eval/local_runs/wisco_official_tier1_precise_deadline_full_20260809T190821Z/hierarchical/20260809T192256Z_wisco_official_tier1_precise_deadline_full_hierarchical.csv`. **Commands are preserved in** `Documentation/AI_HANDOFF/CLAUDE_TASK_36_FINAL_REPORT.md`; **the input hashes and eligibility checks are in** `eval/local_runs/wisco_official_tier1_precise_deadline_full_20260809T190821Z/analysis_task37_1_clean_reproduction/analysis_manifest.json`.
- **B:** `eval/local_runs/e5large_full_heldout_20260824/results.csv`, **with encoder identity, dimensions, and verified collection counts in** `eval/local_runs/e5large_build_manifest_20260823.json`. **The build manifest establishes collection construction, not every setting used during the subsequent evaluation.**
- **C:** `eval/results/raw_runs/enriched_catalogue_heldout_20260824/20260824T113739Z_flat.csv`, **with the later manifest** `eval/results/raw_runs/enriched_catalogue_heldout_20260824/manifests/manifest_enriched_catalogue_heldout_20260824.jsonl`.
- **D:** `eval/results/raw_runs/enriched_e5large_heldout_20260824/20260824T123941Z_flat.csv`, **with the later manifest** `eval/results/raw_runs/enriched_e5large_heldout_20260824/manifests/manifest_enriched_e5large_heldout_20260824.jsonl`.

A-flat/A-hier CSVs record 280ad76; C records 3d10925; D records b43f845. The C/D manifests were generated on 25 August 2026 and record 48f8df6. These are separate historical provenance fields, not the source revision of the present thesis. B’s reduced CSV stores no commit identifier.

D’s CSV incorrectly names the small encoder, while its method label and later manifest select the enriched large-encoder profile. Moreover, the profile is absent from the catalogue registry at the CSV-recorded commit b43f845. That commit marker consequently does not establish a complete snapshot of the code executed for D; the preserved records do not resolve the discrepancy. A subsequent replication should archive the complete executed source, resolved encoder revision, catalogue hashes, dependencies, and command alongside predictions.

F.3 Recorded Computational Environment

The C/D manifests describe Windows 10, Python 3.11.9, an Intel processor reported as *Intel64 Family 6 Model 154 Stepping 4, GenuineIntel*, 12 logical processors, and 7.71 GiB RAM. No CUDA device is detected. Recorded libraries include sentence-transformers 5.2.3, PyTorch 2.10.0, transformers 4.46.3, and qdrant-client 1.17.0. The processor’s commercial model name is not recorded.

These manifests were generated after the executions and describe their archived environment record. A and B lack an equivalent complete environment manifest. Their hardware and dependency configuration cannot be assumed identical to C/D merely because the experiments were conducted on a local machine. Cross-run latency differences are descriptive measurements, not a fully controlled hardware comparison.

F.4 Synthetic-Generation Record

The 449-row source export `eval/results/synthetic_isic_iscedf_benchmark/synthetic_isic_iscedf_benchmark_20260827T021738Z.csv` records `groq/openai/gpt-oss-120b` as `generation_model` and `v2_2026-08-27_multilingual` as `prompt_version` for every row. It also stores the intended category and a source-definition hash. The corresponding measured comparison is `eval/results/synthetic_isic_iscedf_benchmark/synthetic_eval_results_20260827T031325Z.csv`.

The examined generator, `eval/generate_synthetic_isic_iscedf_benchmark.py`, constructs the selected model through `get_llm_strict()` at temperature 0.8. This is a source-configuration observation; the generation CSV does not record temperature, sampling settings, provider model revision, or per-call request metadata. Its model and prompt strings should not therefore be treated as a complete historical execution trace. The two refusal descriptions in the original measured corpus and the distinction between generating labels and independent expert labels remain as described in Chapters 5 and 6.

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